# Smartphone-Only Estimation of Elevator Vertical Displacement with Calibrated Confidence Intervals

EYAL YAKIR, TBD, TBD

Knowing which floor a smartphone user is on remains an open problem on devices that lack a pressure sensor, and the single most common floor-change event — the elevator ride — is exactly where the three workhorses of indoor localisation each break down. Pedestrian dead reckoning and step counting estimate horizontal trajectory; the cabin removes the steps and the accelerometer cannot be integrated vertically without unbounded drift. Wi-Fi fingerprinting demands a per-floor survey, degrades whenever the radio environment drifts, and is shielded by the cabin itself. Barometric altimetry resolves a floor in principle but assumes a pressure sensor that entry-level phones omit, and demands re-referencing against weather drift on phones that carry one.

We present a smartphone-only pipeline that estimates the signed vertical displacement $\Delta h$ of every elevator ride a user takes and attaches a calibrated 90 % confidence interval to it, using the phone's accelerometer alone. The pipeline rests on one observation. The mechanical limits of an elevator cabin force the gravity-removed acceleration trace a phone records during a ride into a small parametric family of basis functions. We decompose every candidate ride against this family in two stages — segmentation of hours-long inertial streams into discrete up / down ride intervals, and per-ride prediction of $\Delta h$ with a per-ride uncertainty. Each interval is distribution-free with finite-sample coverage guarantees, adapts to ride duration, gravity-vector stability, and fit quality, and remains valid under both white sensor noise and the coloured (drift- and vibration-driven) noise that dominates real hand-held recordings.

We evaluate on **[TODO: 150]** experiments spanning **[TODO: TBD]** elevators in **[TODO: TBD]** buildings (residential, office, hotel), with a strict held-out blind test. Median CI half-width on the held-out test stays below **[TODO: $\pm 0.9$ m]** on short rides at 90 % coverage, an improvement of **[TODO: TBD %]** over the dataset's prior best. We open-source the pipeline, dataset, and evaluation harness.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing**; • **Computing methodologies** → *Sensor data processing*.

Additional Key Words and Phrases: smartphone sensing, inertial measurement, elevator, vertical localization, conformal prediction, uncertainty quantification, indoor positioning, matched filter

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## 1 Introduction

Indoor location services place a smartphone within metres of the right room on the horizontal plane, using Wi-Fi fingerprinting, BLE beacons, and visual pedestrian dead reckoning [9]. The vertical axis lags behind. Knowing the user's floor matters for emergency dispatch, which routes responders to a floor rather than to a building entrance. It matters for venue-aware advertising and indoor delivery in malls and office towers, and for augmented-reality experiences anchored to a specific floor of a museum or conference centre.

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Three structural limitations make floor estimation harder than its horizontal sibling. *(i)* Step counting and pedestrian dead reckoning estimate horizontal trajectory and report nothing about height [5]. *(ii)* Wi-Fi fingerprinting separates floors only where the survey covered every floor and the radio environment has not drifted since [2]. *(iii)* Barometric altimetry resolves a floor in principle, but the pressure sensor is absent on entry-level phones and demands re-referencing against weather drift on phones that carry one [12]. The single most common floor-change event — the elevator ride — sits in the regime where every horizontal-localisation signal degrades at once.

Elevators are not arbitrary motion. We found that the mechanical limits of the cabin force the gravity-removed acceleration trace a phone records during a ride into a small parametric family of expected functions. Section 2 characterises this family precisely; what matters in the introduction is that the family is small enough to enumerate and structured enough to fit a candidate ride against. Every claim in this paper rests on it.

The family turns a hard goal into a tractable one. Given an inertial recording of a single elevator ride, we recover the ride's signed vertical displacement $\Delta h$ together with a calibrated 90 % confidence interval. Recovery reduces to fitting a candidate ride against the family, which converts a smartphone into a per-ride vertical-displacement sensor that needs no barometer at inference time, no Wi-Fi survey, and no per-building calibration.

Our pipeline has two stages. Segmentation extracts ride intervals from a continuous, possibly hours-long, inertial stream. Prediction returns $\Delta h$ for each interval together with a per-ride uncertainty. Both stages read the phone's accelerometer alone. The per-ride confidence interval is distribution-free, carries finite-sample coverage guarantees, and remains valid under both white sensor noise and the coloured (drift- and vibration-driven) noise that dominates real hand-held recordings.

We evaluate on **[TODO: 150]** experiments spanning **[TODO: TBD]** elevators across **[TODO: TBD]** buildings (residential, office, hotel), with a strict held-out blind test. The system retains its nominal 90 % coverage on the blind test. The median CI half-width stays below **[TODO: ±0.9 m]** on short rides, and per-bin error improves on the dataset's prior best by **[TODO: TBD %]**. Section 5 reports the full coverage and error statistics.

*Contributions.*

(1) **Per-ride displacement with calibrated coverage.** We deliver 90 %, distribution-free, finite-sample confidence intervals on $\Delta h$ from accelerometer data alone, valid under both white sensor noise and the coloured noise that dominates hand-held recordings.
(2) **Family-of-basis-functions framework.** We characterise the family of expected gravity-removed acceleration traces for an elevator ride from the cabin's mechanical limits, and use the same family as the substrate for both segmentation and per-ride prediction.
(3) **Dataset and harness.** We release the multi-building dataset (**[TODO: 150]** experiments, multiple phones per ride, barometer ground-truth pipeline), the full evaluation harness, and the operator UI used to triage rides.

*Roadmap.* Section 2 characterises the elevator's mechanical structure, the physical limits it imposes on a ride, and the family of expected acceleration functions those limits force. Section 4 presents the two-stage algorithm: it segments an inertial stream into rides and predicts each ride's $\Delta h$ with a calibrated interval. Section 3 describes the dataset and its ground truth, and Section 5 reports the train/test split and the results. Sections 6 and 7 discuss limitations and conclude.

## 2 Background: Elevator Kinematics

This section derives the family of acceleration functions an elevator ride can produce on a smartphone. We start from the cabin's mechanical parts and the physical constraints they impose (§2.1), translate those constraints into bounds on jerk, acceleration, and velocity (§2.2), and then write down the seven-segment S-curve that the bounds force every ride to follow (§2.3). We close with the closed-form relations between machine parameters and the time constants of the ride (§2.4) and the parametric family that this paper's segmentation and prediction stages fit (§2.5); the heavy algebra lives in Appendix A.

### 2.1 How a passenger elevator works

A passenger elevator is, mechanically, a single moving piece: a cabin (the *car*) hanging on a steel rope that runs up over a sheave at the top of the shaft and down the other side to a counterweight (Figure 1). A motor turns the sheave; the cabin and counterweight rise and fall in opposite directions. The counterweight is sized to match the empty car plus roughly half the rated payload, so the motor only has to drive the small mass difference and the inertia of the moving parts. A brake clamps the sheave shaft whenever the motor is unpowered or anything goes wrong. Three machine pieces decide how the cabin can move: the rope (it stretches), the motor (it has a torque ceiling), and the brake (it has a maximum stopping distance fixed by the headroom $h_1$). The next subsection turns each into a hard limit on the cabin's motion.

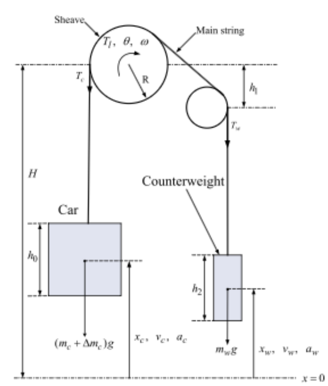



Fig. 1. A traction elevator at a glance. A sheave of radius $R$ couples the car ($m_c$+$\Delta m_c$) to a counterweight ($m_w$). The motor delivers torque $T_j$ at angular velocity $\omega$; rope tensions $T_v$ and $T_w$ load the sheave. $H$ is the hoistway height; $h_0$ and $h_2$ are the cabin and counterweight heights; $h_1$ is the headroom budget above the sheave that the brake must respect.

### 2.2 Physical limitations and what they bound

Four mechanical realities turn into four hard limits on the cabin's motion. Each is the source of one of the parameters our pipeline fits.

*The rope sets the maximum jerk.* The main rope stretches. If the motor changes torque too quickly, the cabin starts to bounce on the rope: the passenger feels it as a thump, and the rope itself fatigues over time. Drive controllers therefore ramp torque through a slew-rate limit [10]. Slew-rate-limiting the torque is the same as putting a ceiling on the jerk $\dddot{h}$ the cabin feels. The ride-comfort standard ISO 18738 pins this at $j_{\max} \approx 1.0–1.6\,\mathrm{m/s^3}$ for passenger elevators [6].

*The motor sets the maximum acceleration.* At constant velocity the motor only has to fight friction and the small mass mismatch between cabin and counterweight. When it accelerates, it has to push the entire moving mass. The motor has a torque ceiling, so cabin acceleration is also capped: $a_{\max} \approx 0.8–1.5\,\mathrm{m/s^2}$ on commercial traction elevators. Older geared machines sit at the low end; newer gearless ones at the high end.

*The brake sets the maximum cruise velocity.* If anything goes wrong — a slack rope, an overspeed sensor trip, loss of power — the cabin has to stop before it hits the top of the shaft. The brake decelerates the cabin at some rate $a_{\text{brake}}$ and has at most the headroom $h_1$ above the top floor to do it in. Energy says $v_{\max}^2 \le 2\, a_{\text{brake}}\, h_1$, which in practice limits cruise velocity to $v_{\max} \approx 1.0$–$2.5$ m/s in low- and mid-rise buildings (CIBSE Guide D [4]). Faster cabins exist in skyscrapers, but they require taller headroom and shaft-mounted buffers.

*Only the height changes from ride to ride.* Once the elevator is built, $j_{\max}$, $a_{\max}$, and $v_{\max}$ are fixed for the building — often fixed for the manufacturer and service class. The only thing that changes ride to ride is the floor-to-floor height the cabin has to cover, which we write $H$ (metres, signed for direction). Putting the three limits together, the cabin obeys

$$\left|\dddot{h}(t)\right| \le j_{\max}, \qquad \left|\ddot{h}(t)\right| \le a_{\max}, \qquad \left|\dot{h}(t)\right| \le v_{\max}, \tag{1}$$

where $\ddot{h} = a$ is the gravity-removed vertical acceleration the phone measures.

### 2.3 Why the ride is an S-curve

*The optimisation problem.* The drive solves a simple problem: move the cabin $H$ metres, starting and ending at rest, in the shortest possible time, without ever breaking the three limits in (1). Write the cabin's altitude as $h(t)$, its velocity as $v(t) = \dot{h}(t)$, its acceleration as $a(t) = \ddot{h}(t)$, and the controller's command as the jerk $j(t) = \dddot{h}(t)$. The controller picks $j(t)$; everything else follows by integration. Formally we solve

$$\begin{aligned} \underset{T,\, j(\cdot)}{\text{minimize}} \quad & T \\ \text{subject to} \quad & h(0) = 0, \quad h(T) = H, \quad v(0) = v(T) = 0, \\ & |j(t)| \le j_{\max}, \quad |a(t)| \le a_{\max}, \quad |v(t)| \le v_{\max}, \\ & \forall\, t \in [0, T]. \end{aligned} \tag{2}$$

*The result.* Pontryagin's maximum principle, applied to (2), forces the optimal command to be *bang-bang* on the highest-derivative input. At every instant, the optimal jerk takes only three values,

$$j^*(t) \;\in\; \{-j_{\max},\ 0,\ +j_{\max}\}, \tag{3}$$

with $j^* = 0$ holding the cabin against $|a| = a_{\max}$ or $|v| = v_{\max}$ when one of the lower-order limits is itself saturated. Intuitively, any moment spent at an interior value $|j| < j_{\max}$ is wasted: the cabin could have been pushing harder and finished sooner. Appendix A.1 sketches the derivation; we use (3) below.

*The S-curve and its three regimes.* Stitching the bang-bang jerk command (3) together with the three limits yields the canonical *S-curve* acceleration profile. The floor-to-floor height $H$ alone fixes how many of the limits a ride saturates, and so splits every ride into one of three regimes — long, medium, or short — shown side by side in Figure 2.

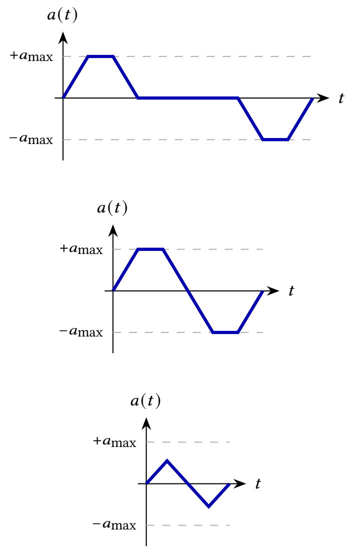
### 2.4 From machine parameters to ride-time constants

The bang-bang structure pins every phase of the ride to a closed-form duration in the machine parameters and the floor-to-floor height $H$. Three durations describe any ride:

$$T_j = \frac{a_{\max}}{j_{\max}}, \tag{4}$$

$$T_a = \max\left(0,\ \frac{v_{\max}}{a_{\max}} - T_j\right), \tag{5}$$

$$T_v = \max\left(0,\ \frac{H}{v_{\max}} - 2T_j - T_a\right), \tag{6}$$




**Long ride — 7 segments.**
$T_v > 0, \quad T_a > 0$
All three limits saturate; a flat $v_{\max}$ cruise splits the two trapezoidal lobes.

**Medium ride — 5 segments.**
$T_v = 0, \quad T_a > 0$
Jerk and acceleration saturate but the cabin never reaches $v_{\max}$; the two lobes touch at the peak-velocity instant.

**Short ride — 3 segments.**
$T_v = T_a = 0$
Only jerk saturates; the two lobes are triangles that meet, with peak $|a| < a_{\max}$.

Fig. 2. The three S-curve regimes of the time-optimal acceleration profile $a(t)$, all produced by the bang-bang jerk command (3) and selected by the floor-to-floor height $H$. Each row pairs the profile with the time-constant condition that defines its regime; longer $H$ saturates more limits. Closed forms for $T_j$, $T_a$, and $T_v$ follow in §2.4.

and they sum over the seven phases to the total ride duration

$$T_{\text{ride}} = 4T_j + 2T_a + T_v = \frac{H}{v_{\max}} + \frac{v_{\max}}{a_{\max}} + \frac{a_{\max}}{j_{\max}} \quad \text{(trapezoidal regime).} \tag{7}$$

Figure 3 locates the four durations on a single ride. The jerk ramp $T_j$ and the acceleration plateau $T_a$ are constants of the building. $T_j$ depends only on the machine — about 0.77 s for a typical lift ($a_{\max} = 1.0\,\text{m/s}^2$, $j_{\max} = 1.3\,\text{m/s}^3$) — and $T_a$ vanishes on rides too short to reach $v_{\max}$. Only the cruise $T_v$ scales with ride length, growing as $H/v_{\max}$. Equation (7) is the inversion handle later sections use to recover $H$ jointly with the machine parameters; Appendix A derives all four constants from the bounds in (1).

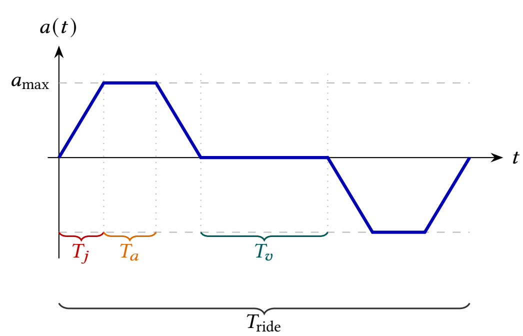



Fig. 3. The four ride-time constants on the acceleration trace $a(t)$ of a trapezoidal ride: jerk ramp $T_j$, the $a_{\max}$ plateau $T_a$, the $v_{\max}$ cruise $T_v$ (the flat $a = 0$ gap between lobes), and the total ride $T_{\text{ride}}$.

### 2.5 The trapezoid basis function and the parametric family

Combining (1)–(7), every ride is fixed by four parameters,

$$\boxed{\theta = (j_{\max}, a_{\max}, v_{\max}, H)}, \tag{8}$$

three constants of the building and the ride-specific height $H$.

*The trapezoid basis function.* Each acceleration lobe of the S-curve is a trapezoid in time — a ramp up, a flat top, and a ramp down. We normalise it to the trapezoid basis function

$$\tau_{W,f}(t) = \begin{cases} 1 & |t| \le fW, \\ \dfrac{W - |t|}{(1-f)\,W} & fW < |t| \le W, \\ 0 & |t| > W, \end{cases} \tag{9}$$

of *half-width* $W$ and *flat fraction* $f \in [0,1]$: a unit plateau over $[-fW, fW]$ joined to zero by two linear ramps (Figure 4).

*Mapping to the machine parameters.* The first lobe spans $2T_j + T_a$, so the machine parameters fix $(W, f)$ exactly:

$$W = T_j + \tfrac{1}{2}T_a = \frac{a_{\max}}{2\,j_{\max}} + \tfrac{1}{2}\max\!\Big(0,\ \tfrac{v_{\max}}{a_{\max}} - T_j\Big), \tag{10}$$

$$f = \frac{T_a}{T_a + 2T_j} = \frac{\max\big(0,\ v_{\max}/a_{\max} - T_j\big)}{\max\big(0,\ v_{\max}/a_{\max} - T_j\big) + 2T_j}. \tag{11}$$

A short hop that never reaches $v_{\max}$ has $T_a = 0$ and $f = 0$, a pure triangle; a long ride has $T_a \gg T_j$ and $f \to 1$, a near-rectangular plateau. The two lobes pull apart — their centroid spacing $\Delta t_c$ exceeds $2W$ — only once a $v_{\max}$ cruise opens between them, the 7-segment regime of §2.3.

*The parametric family.* Every ride is therefore a pair of opposite-sign trapezoids of shared shape,

$$a_\theta(t) \;=\; s\,A\,\tau_{W,f}(t - t_{c,1}) \;-\; s\,A\,\tau_{W,f}(t - t_{c,2}), \tag{12}$$

with amplitude $A = a_{\max}$, direction $s \in \{+1, -1\}$, and centroid spacing $\Delta t_c = t_{c,2} - t_{c,1} = H/v_{\max}$ (Appendix A).

Three facts about this family $\mathcal{F}_{\text{ride}}$ drive the rest of the paper. *(i)* Three of its parameters are constants of the building, so a ride differs from the building's other rides only in the height $H$ — the single quantity the predictor of §4 recovers. *(ii)* The two lobes carry equal and opposite area, so $\int_0^{T_{\text{ride}}} a_\theta(t)\,dt = 0$ — the zero-mean anchor for any ZUPT-style predictor. *(iii)* The lobe spacing $\Delta t_c = H/v_{\max}$ is a scale-free estimator of $H$, so the matched-filter detector needs no per-building amplitude calibration.

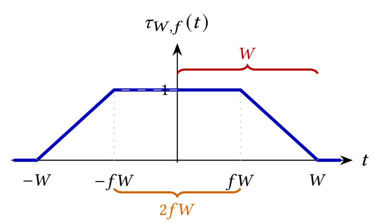



Fig. 4. The trapezoid basis function $\tau_{W,f}(t)$ of (9).

### 2.6 The smartphone inertial sensors

The pipeline reads two of the phone's inertial sensors — the accelerometer and the gyroscope — and one obstacle separates their raw signal from the vertical acceleration a ride produces: the phone's unknown, time-varying orientation.

*The accelerometer.* Every phone carries a six-axis MEMS IMU [3]. Its `TYPE_LINEAR_ACCELERATION` stream reports the gravity-removed acceleration as a three-vector $\mathbf{a} = (a_x, a_y, a_z)$ along the phone's *body axes* — the $x, y, z$ fixed to the case (Figure 5). The sensor-hub rate varies by device, so we resample every recording to a uniform 100 Hz grid ($\Delta t = 10$ ms). The noise floor is white, but a hand-held phone

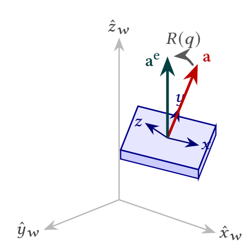



Fig. 5. The phone *records* $\mathbf{a} = (a_x, a_y, a_z)$ in its tilted body frame; we *want* the earth-frame $\mathbf{a}^{\mathrm{e}} = (a_x, a_y, a_z)_{\mathrm{e}} = R(q)\,\mathbf{a}$, whose vertical channel is the ride signal.

adds coloured drift and 1–50 Hz vibration, so later sections fit the noise model on real ride residuals.

*The unknown vertical.* An elevator ride is purely vertical: the cabin accelerates along the world up-axis $\hat{z}_w$, and that one scalar $a(t)$ is all the pipeline wants. The accelerometer, though, measures along the phone's body axes, and a hand-held phone tilts arbitrarily. An unknown rotation $R$ separates the two frames, so the vertical signal spreads across all three channels, $\mathbf{a} = R^{\top} a\, \hat{z}_w$ — the body channel $a_z$ is not the world vertical. Recovering $a(t)$ means recovering which body direction points up at every instant. Gravity in the stationary windows around a ride fixes that direction only at rest; a hand carried through a ride rotates, so the direction must be tracked through the ride itself.

*Reconstructing the world frame.* A Mahony passive complementary filter [11] tracks the body-to-world orientation, seeded from the opening stationary window. The orientation is a rotation matrix $R \in SO(3)$ ($R^{\top}R = I$, $\det R = 1$) that maps body-frame vectors to the world frame; we carry it as a unit quaternion $q = (\cos\frac{\theta}{2},\ \hat{\mathbf{n}}\sin\frac{\theta}{2})$, a four-number, singularity-free encoding of a rotation by angle $\theta$ (radians) about the unit axis $\hat{\mathbf{n}}$, and write the induced matrix $R(q)$. The gyroscope $\boldsymbol{\omega}$ propagates $q$ and the accelerometer (gravity retained, $\hat{\mathbf{a}} = \mathbf{a}/\|\mathbf{a}\|$) corrects it toward vertical.

$$\hat{\mathbf{v}} = R(q)^{\top}\hat{z}_w, \qquad \mathbf{e} = \hat{\mathbf{a}} \times \hat{\mathbf{v}}, \tag{13}$$

$$w = \exp\!\Big(-\tfrac{1}{2}\big((\|\mathbf{a}\| - g)/\tau g\big)^2\Big), \qquad \dot{\mathbf{b}} = -k_I\, w\, \mathbf{e}, \tag{14}$$

$$\dot{q} = \tfrac{1}{2}\, q \otimes \big[\,0,\ \boldsymbol{\omega} - \mathbf{b} + k_P\, w\, \mathbf{e}\,\big]^{\top}, \tag{15}$$

$$\mathbf{a}^{\mathrm{e}} = \big(a^{\mathrm{e}}_x, a^{\mathrm{e}}_y, a^{\mathrm{e}}_z\big) = R(q)\,\mathbf{a}, \qquad a = a^{\mathrm{e}}_z - g. \tag{16}$$

The proportional term $k_P w\mathbf{e}$ pulls $q$ toward gravity, the integral term learns the gyro bias $\mathbf{b}$, and the trust weight $w \to 0$ freezes the correction when $\|\mathbf{a}\| \neq g$, so ride acceleration is never read as tilt ($k_P = 1$, $k_I = 0.3$). With no gyroscope, $q$ holds at its seed and the vertical channel falls back to the rest-frame magnitude $\|\mathbf{a}\| - g$, which the mid-ride tilt corrupts. Tracking $q$ instead cuts the per-ride error close to threefold overall — halving the mean $\Delta h$ error and shrinking the RMSE tail 4–6× (Appendix E.2).

## 3 Dataset and Ground Truth

The dataset is a field campaign of **[TODO: ≈150]** elevator rides recorded across **[TODO: 7]** elevators in **[TODO: 6]** buildings in central Israel (Table 5). The buildings span residential, office, and hotel cabins, each with a different cabin geometry, motor, and floor-to-floor height, so no estimator can overfit a single machine. Four phones, carried together, recorded every ride, so one physical event produced four heterogeneous accelerometer traces — different MEMS parts, sampling jitter, and noise floors. At least one of the four phones carries a barometric pressure sensor, which serves as the ground-truth device even when the phone under test has none.

Every ride's signed displacement $\Delta h$ comes from the barometer rather than from hand-labelling. We invert the pressure stream to altitude through the ISA barometric formula; a consumer barometer resolves altitude to roughly 10 cm, an order of magnitude finer than the ~3 m inter-floor pitch the pipeline must separate. Low-pass filtering and differencing the same altitude trace yields the `up`, `down`, and `outside` interval labels that cut each session into discrete rides. A phone without a barometer borrows the pressure stream of a

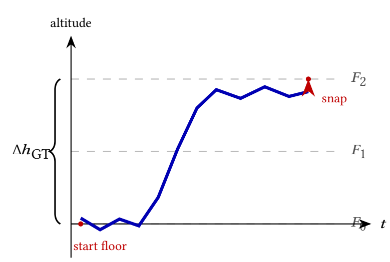



Fig. 6. Snapping the drifting barometer altitude (blue) to the nearest *gramushka* floor line (dashed) yields a drift-free ground-truth $\Delta h$.

co-carried device, aligned to it by accelerometer cross-correlation (Appendix B).

Raw barometric altitude drifts with weather and building airflow, so we snap it to the building itself. Each building contributes a *gramushka* table — the floor elevations read from its accordion-folded architectural drawing (Figure 6). We anchor every session at its known start floor and integrate the barometer altitude across each detected ride. Snapping the ride endpoint to the nearest table floor makes the ground-truth $\Delta h$ a true floor-to-floor displacement. The pipeline flags any snap whose distance exceeds **[TODO: TBD]** m for manual review rather than forcing it, and the one elevator without a drawing falls back to the drift-corrected barometer $\Delta h$ (Appendix B).

## 4 Algorithm

The pipeline converts a continuous, hours-long accelerometer stream into one signed displacement $\Delta h$ per elevator ride, each carrying a calibrated 90 % confidence interval, in two stages (Figure 7). Segmentation slices the stream into discrete ride intervals (§4.1); prediction fits each interval for its $\Delta h$ and uncertainty (§4.2). Both stages reuse the trapezoid family of §2.5 directly: a ride is an opposite-sign pair of trapezoid lobes (12), so the matched filter that finds a ride is the matched filter that measures it.

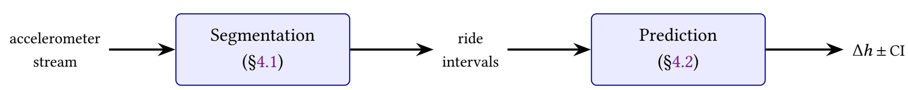



Fig. 7. The two-stage pipeline. Segmentation turns a continuous accelerometer stream into discrete ride intervals; prediction turns each interval into a signed displacement $\Delta h$ with a calibrated confidence interval. Both stages fit the trapezoid ride family of §2.5.

### 4.1 Segmentation

Segmentation cuts an hours-long accelerometer stream into discrete ride intervals, each tagged up or down. A ride is an opposite-sign pair of trapezoid lobes (12), so segmentation detects candidate lobes (§4.1.1) and then pairs them into rides (§4.1.2).

*4.1.1 Detecting candidate lobes.* The pipeline projects the three-axis accelerometer onto the session's gravity direction (§2.6) and smooths it into a signed vertical trace $a(t)$. A least-squares matched filter scores $a(t)$ against every lobe template $\tau_{W,f}$ of the family $\mathcal{F}_{\text{ride}}$ (§2.5): at each sample $i$, fitting one amplitude to the length-$2W$ window $a_{[i]}$ centred on $i$ gives

$$\hat{A}_{W,f}(i) = \frac{\langle a_{[i]},\, \tau_{W,f} \rangle}{\|\tau_{W,f}\|^2}, \qquad R^2_{W,f}(i) = 1 - \frac{\|a_{[i]} - \hat{A}_{W,f}(i)\, \tau_{W,f}\|^2}{\|a_{[i]}\|^2}. \tag{17}$$

Collapsing the $(W, f)$ grid by amplitude sign yields, per sample,

$$R^2(i) = \max_{W,f} R^2_{W,f}(i), \qquad R^2_{\pm}(i) = \max_{\pm \hat{A}_{W,f}(i) > 0} R^2_{W,f}(i), \tag{18}$$

where a high $R^2_+(i)$ marks a take-off lobe and a high $R^2_-(i)$ a landing lobe. A candidate lobe is a strict local maximum of $R^2_+$ or $R^2_-$ that clears an amplitude and a shape floor,

$$|\hat{A}(i)| \geq a_{\min}, \qquad R^2(i) \geq r_{\min}, \tag{19}$$

with $\hat{A}(i)$ the amplitude at the best grid cell; non-maximum suppression then drops near-duplicates within $\delta_{\text{nms}}$ and keeps same-sign lobes at least $\Delta_{\text{ss}}$ apart (Figure 8). The grid bounds (10)–(11) and the thresholds are calibrated on the training set (Appendix C).

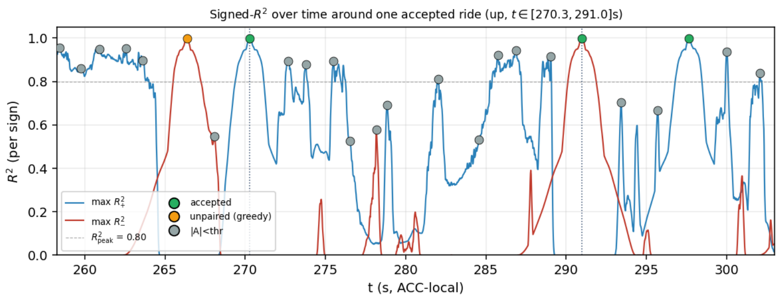



Fig. 8. Signed score traces $R^2_+(t)$ (take-off, blue) and $R^2_-(t)$ (landing, red) around one ride. Local maxima are candidate lobes; dot colour marks the first stage to reject each peak, leaving the committed take-off/landing pair (green).

#### 4.1.2 *Pairing candidate lobes into rides.*

Each candidate lobe $i$ admits an opposite-sign partner set inside the ride-duration window,

$$\mathcal{P}(i) = \left\{ j : \operatorname{sign} \hat{A}(j) = -\operatorname{sign} \hat{A}(i), \quad t_{\min} \le |t_j - t_i| \le t_{\max} \right\}. \tag{20}$$

A genuine ride's two lobes are cut from one cabin profile, so each pair $(i, j)$ is fitted with a single template and a single shared amplitude. With $a_k$ the $k$th lobe window, $P_k = \|a_k\|^2$, $u_k = s_k \langle a_k, \tau_{W,f} \rangle$ its sign-corrected correlation, and $N = \|\tau_{W,f}\|^2$,

$$A^\star(W, f) = \frac{u_1 + u_2}{2N}, \qquad R_k^2(W, f) = 1 - \frac{P_k - 2A^\star u_k + (A^\star)^2 N}{P_k}, \tag{21}$$

and the pair score is $S(i, j) = \max_{W,f} \frac{1}{2}(R_1^2 + R_2^2)$, attained at $(W^\star, f^\star, A^\star)$. These starred values describe one *average* lobe: the fit assumes take-off and landing share a shape, so $A^\star$ averages the two single-lobe amplitudes. A high $S$ means both lobes trace out that same trapezoid. The pair is admissible when

$$S(i, j) \ge r_{\text{pair}}, \quad A^\star \ge a_{\text{pair}}, \quad E(i, j) \ge e_{\min}, \quad \text{RMS}_{\text{mid}} \le \rho\, A^\star. \tag{22}$$

The grid energy $E$ averages the clamped joint $R^2$ over every cell of the $(W, f)$ template grid $G$,

$$E(i, j) = \frac{1}{|G|} \sum_{(W,f) \in G} \max\!\Big(0,\ \tfrac{1}{2}\big(R_1^2(W, f) + R_2^2(W, f)\big)\Big), \tag{23}$$

so a real ride — which lights a broad band of cells, not one — clears the floor $e_{\min}$. The cruise term $\text{RMS}_{\text{mid}}$ is the trace RMS on the inter-lobe stretch, near zero while the cabin holds constant velocity. A greedy resolver ranks admissible pairs by

$$\operatorname{rank}(i, j) = S(i, j) - \lambda\, |t_j - t_i| \tag{24}$$

and commits them highest-first, skipping any that reuse a lobe or overlap a committed interval. The penalty $\lambda$ defeats the *super pair* — a take-off matched to a far-later landing, which scores like a true ride but spans far longer. The committed pairs are the ride intervals passed to prediction.

**Appendix C explains and illustrates every detection and pair-filter threshold.**

### 4.2 Prediction

Prediction takes one ride interval and returns the signed displacement $\Delta h$, a per-ride standard deviation $\sigma$, a calibrated 90 % confidence interval, and an `accepted` flag. Displacement follows from one fact: an accelerometer measures acceleration, and displacement is its double integral — integrate the vertical trace $a(t)$ once for velocity, twice for $\Delta h$. The front end recovers that signed vertical trace from the three-axis accelerometer by the gravity projection of §2.6.

Two accelerometer-only algorithms run that double integral by different routes. ZUPT double integration (§4.2.1) integrates the measured $a(t)$ directly, anchored by the cabin resting at both ends of the ride. The trapezoid pulse-pair fit (§4.2.2) integrates a model instead: it constructs the ride's theoretical acceleration signal — the trapezoid pulse pair of §2.5 — and integrates it in closed form, reading $\Delta h$ off a formula in the fitted shape. Section 5 compares them head to head.

Both feed three shared stages: a per-ride error model (§4.2.3), a conformal layer that calibrates $\sigma$ into the interval (§4.2.4), and a quality filter that sets the `accepted` flag (§4.2.5). The trapezoid fit borrows only ZUPT's cruise-velocity scale as an amplitude anchor, never its displacement, so the two stay independent estimators and the comparison of §5 stays meaningful.

*4.2.1 ZUPT double integration.* ZUPT recovers $\Delta h$ by double-integrating the measured vertical acceleration under a zero-velocity boundary at both ends of the ride,

$$v(t) = \int_0^t a(\tau)\,d\tau \;-\; \frac{t}{T}\int_0^T a(\tau)\,d\tau, \qquad \Delta h = \int_0^T v(t)\,dt, \tag{25}$$

over a ride of duration $T$.

The boundary assumption anchors the integral. A ride begins and ends with the cabin at rest, so $v(0) = v(T) = 0$. The first integral of (25) enforces $v(0) = 0$ by construction; the linear ramp term subtracts the integrated-noise drift that would otherwise leave $v(T) \neq 0$ — the zero-velocity update that names the method.

ZUPT integrates only the moving span: the pipeline thresholds the vertical trace at $|a(t)| = 0.10\,\mathrm{m/s^2}$ and zeroes the quiet pre- and post-roll, so noise outside the ride never enters the integral. The count $N$ of active samples sets the scale of the error model (§4.2.3).

*4.2.2 Trapezoid pulse-pair fit.* The trapezoid fit recovers $\Delta h$ from the *shape* of the acceleration trace. The ride's theoretical signal is the S-curve of §2.3, set by three machine parameters $(j_{\max}, a_{\max}, v_{\max})$ besides the height. Fitting all three to a hand-held recording is ill-conditioned: the jerk ramps occupy too small a fraction of the ride to constrain them. The fit instead collapses each acceleration lobe to the trapezoid $\tau_{W,f}$ the detector already uses, a two-parameter shape the bang-bang jerk command makes exact (§2.5).

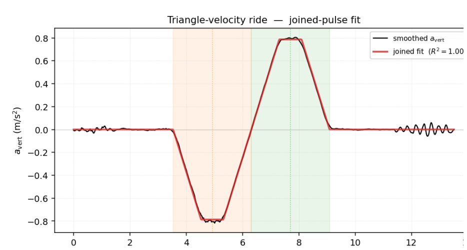



Fig. 9. The reconstructed signal $a_\theta(t)$ (red) overlaid on the smoothed vertical trace (black) for one ride: two opposite-sign trapezoid lobes centred at $t_{c,1}$ and $t_{c,2}$.

The segmenter (§4.1.2) already returns the fitted parameters $(W, f, A, s, t_{c,1}, t_{c,2})$ that reconstruct the ride's acceleration as the opposite-sign trapezoid pair

$$a_\theta(t) \;=\; sA\,\tau_{W,f}(t - t_{c,1}) \;-\; sA\,\tau_{W,f}(t - t_{c,2}), \tag{26}$$

the estimated ride signal (Figure 9). Double-integrating $a_\theta(t)$ gives $\Delta h$ in closed form,

$$\Delta h \;=\; s\,A\,W\,(1+f)\,\Delta t_c, \qquad \Delta t_c = t_{c,2} - t_{c,1}; \tag{27}$$

Appendix D works through the integral and the joined-pulse limit.

*4.2.3 Theoretical error analysis.* Every prediction carries a per-ride standard deviation $\sigma$ derived from its own fit, not a constant. The two algorithms reach that $\sigma$ from different error sources.

*ZUPT.*. Double integration accumulates sensor noise. Integrating white noise of standard deviation $\sigma_a$ twice over $N$ active samples of spacing $\Delta t$ gives an end-point displacement variance whose root is

$$\sigma_{\text{white}} \;=\; \sigma_a\,\Delta t^2\,\sqrt{N^3/12}, \tag{28}$$

the classic ZUPT result, with the factor $1/12$ crediting the linear drift correction of (25) for removing one degree of freedom. A hand-held phone carries three further error sources, which the total budget adds to the white-noise term in quadrature,

$$\sigma^2_{\text{ZUPT}} \;=\; \sigma^2_{\text{white}} + \sigma^2_{\text{mech}} + \sigma^2_{\text{rel}} + \sigma^2_{\text{proj}}\;: \tag{29}$$

a mechanical-jitter term $\sigma_{\text{mech}}$ for hand and pocket movement, a relative-scale term $\sigma_{\text{rel}}$ that grows with $|\Delta h|$, and a projection penalty $\sigma_{\text{proj}}$ for rides that lack a clean gravity reference. Appendix D derives all four.

*Trapezoid.* The trapezoid budget propagates the matched-filter fit uncertainty through the closed-form height (27) by the delta method,

$$\sigma^2_{\Delta h} \;=\; [W(1{+}f)\,\Delta t_c]^2\,\sigma_A^2 + [A(1{+}f)\,\Delta t_c]^2\,\sigma_W^2 + [AW\,\Delta t_c]^2\,\sigma_f^2 + [AW(1{+}f)]^2\,\sigma^2_{\Delta t_c}, \tag{30}$$

with the per-parameter variances supplied by the matched-filter Cramér–Rao bound (template energy and noise level). Two data-adaptive corrections fit the budget to hand-held data: the effective noise level uses the fit residual rather than the datasheet figure, and the whole budget is divided by the joint $R^2$. Appendix D derives every term.

*4.2.4 Confidence interval.* The per-ride $\sigma$ sets the *shape* of the uncertainty but not its *scale*: nothing so far guarantees that $\pm\sigma$ brackets the true height with any stated probability. Split-conformal calibration [8] supplies the guarantee. On a held-out calibration set the pipeline computes, for every ride $i$, the nonconformity score

$$r_i \;=\; \frac{|\widehat{\Delta h}_i - \Delta h_i|}{\sigma_i}, \tag{31}$$

the prediction error in units of its own $\sigma$, and takes the empirical $(1-\alpha)$ quantile of $\{r_i\}$ as a single multiplier $k$. The reported interval is $\widehat{\Delta h} \pm k\sigma$. By the standard split-conformal argument the interval has distribution-free, finite-sample coverage,

$$\mathbb{P}\Big(\,|\widehat{\Delta h} - \Delta h| \le k\sigma\Big) \;\ge\; 1-\alpha, \tag{32}$$

for exchangeable rides, with no assumption on the error distribution. We set $\alpha = 0.1$ for a 90 % interval.

The same machinery wraps both algorithms, but each calibrates its multiplier separately. ZUPT calibrates $k$ on the clean calibration rides; the trapezoid fit calibrates on the rides that are both clean and quality-accepted, its exchangeable population. The multiplier is the $\lceil(1-\alpha)(n+1)\rceil/n$ empirical quantile of the $n$ scores, floored at 1.645 so a small calibration sample cannot return an interval narrower than the Gaussian 90 % half-width.

The interval is *adaptive* because the multiplier is shared while $\sigma$ is not. A long, cleanly fitted ride and a short, poorly fitted one take the same $k$ but very different $\sigma$, so the interval widens exactly where the error model says the fit is weak — short rides, unstable gravity, low joint $R^2$. Appendix D states the exchangeability assumption and the per-algorithm calibration protocol.

*4.2.5 Quality filter.* The last step attaches the `accepted` flag. Each algorithm runs its own quality filter, but the two share a shape: a handful of interpretable features, a set of hard gates, and an additive penalty score over the graded features. The first breach rejects the ride with a human-readable reason, and the filter rejects whenever any hard gate trips or the penalty score exceeds 6.

The two filters differ in what they inspect. Both check the bracketing-window gravity, the phone's orientation, and the active-motion fraction — any accelerometer estimate is only as good as its vertical axis. ZUPT adds an impact gate and an end-velocity check (Table 7). The trapezoid filter adds fit-quality and geometry checks: the joint $R^2$, the residual autocorrelation, the agreement between the velocity-anchored and matched-filter amplitudes, the residual energy outside the two lobes, and the steadiness of the integrated cruise velocity (Table 8). Appendix D explains and illustrates every check.

The quality filter is a *filter, not a predictor*. It targets the rides where the $\sigma$ model must under-predict the error — a phone rotated mid-ride, an impact that swamps the lobes, a fit with no pulse structure. On those rides no interval width is trustworthy, so the pipeline declines to report rather than report a confident wrong answer. Everywhere else the conformal interval of §4.2.4 carries the coverage guarantee, so the filter passes almost every clean ride and rejects only the pathological.

## 5 Evaluation

The pipeline holds nominal 90 % coverage on a held-out building, places the median per-ride error below half a floor, and emits no phantom rides once the quality filter is in the loop. Three subsections report the result; Appendix E carries the metric definitions, the train/test protocol, and the supporting diagnostic figures.

### 5.1 Segmentation accuracy

| Split | $n_{gt}$ | clean | missed | FP | merged | split | $F1^*$ | IoU-F1@0.5 |
|---|---|---|---|---|---|---|---|---|
| Train (7 sessions) | 415 | 309 | 95 | 29 | 5 | 1 | 0.81 | 0.64 |
| Test (held-out building) | 116 | 98 | 16 | 9 | 1 | 0 | 0.88 | 0.83 |

Table 1. Segmentation on the tuned train pool and on the held-out test building. $F1^*$ is the composite failure-mode score (Appendix E.1); every test-split score sits at or above its train counterpart.

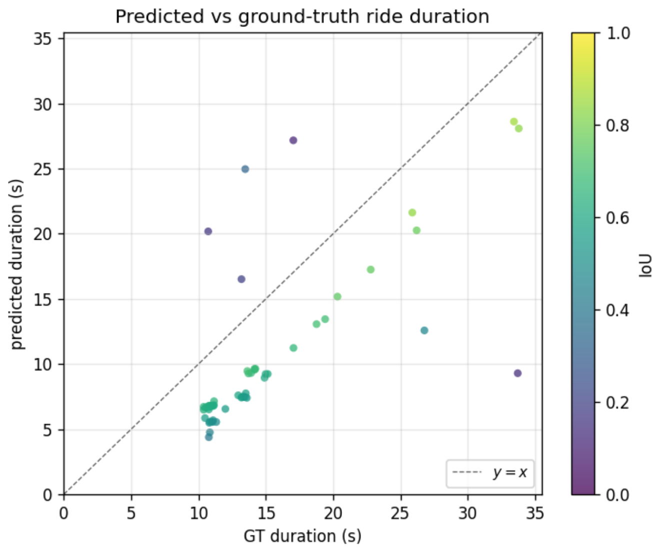



Fig. 10. Predicted ride duration against hand-edited GT duration on every clean match. Points sit systematically below the $y = x$ diagonal: the GT editor pads each labelled interval by 1 s on each side for robustness (§B.1), so the detector's tighter intervals lose a constant $\sim$ 2 s of overlap to the pad. Correcting for the pad lifts the empirical IoU-F1@0.5 of Table 1 from 0.83 towards 0.95 on the test split; the dashed line is the pad-corrected diagonal.

Per-experiment counts, the failure-mode bar chart, and the IoU distribution are in Appendix E.3.

### 5.2 Per-ride prediction

| Algorithm | Split | $n$ | Median (m) | MAE (m) | $\leq$ 1.5 m | $\leq$ 3 m | Coverage | Median CI (m) |
|---|---|---|---|---|---|---|---|---|
| Trapezoid | Train | 478 | 0.24 | 0.97 | 88.5% | 93.3% | 83.5% | 0.94 |
| Trapezoid | Test | 108 | 0.27 | 0.61 | 93.5% | 96.3% | 93.5% | 0.84 |
| ZUPT | Train | 478 | 0.29 | 1.16 | 87.0% | 91.6% | 93.5% | 1.67 |
| ZUPT | Test | 108 | 0.21 | 0.60 | 91.7% | 96.3% | 95.4% | 0.91 |

Table 2. Per-ride prediction on clean-matched intervals, no quality filter. Median error sits below half a floor on both algorithms and both splits; the MAE/RMSE tail comes from a small set of damped recordings the quality filter of Table 3 rejects. Realised coverage at the 90 % nominal holds on the held-out building for both algorithms.

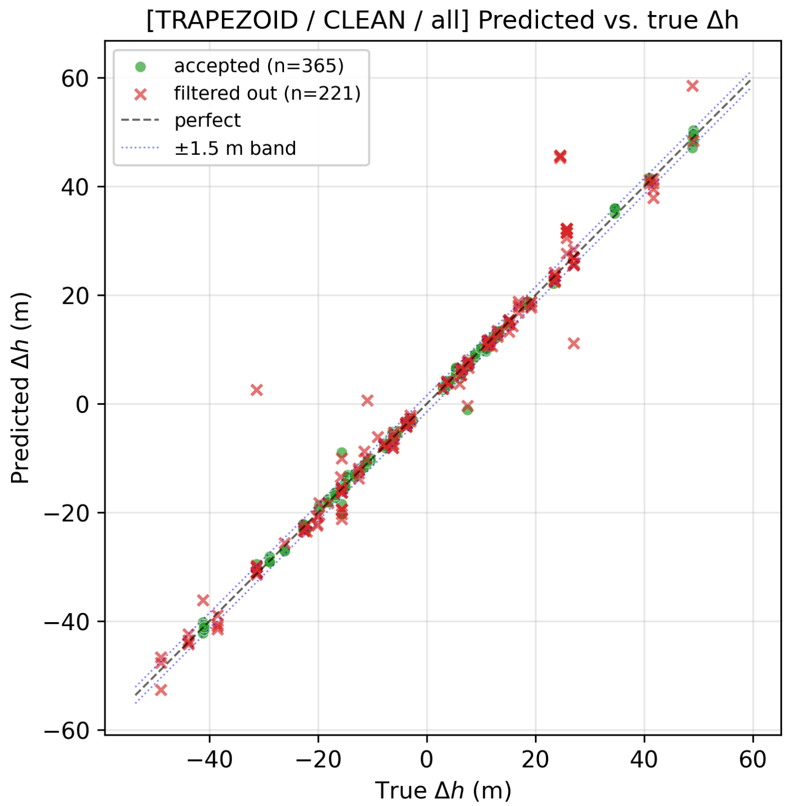



Fig. 11. Predicted versus true $\Delta h$ for the trapezoid fit on every clean ride, pooled across train and test ($n$=586). Points hug the $y$=$x$ diagonal across roughly 50 m of $\Delta h$; green points pass the quality filter, red crosses are rejected, and the dotted lines mark the ±1.5 m one-floor band. Per-algorithm scatter and ZUPT comparison sit in Appendix E.4.

Reliability, coverage by ride duration, and per-experiment error breakdowns are in Appendix E.4.

### 5.3 End-to-end pipeline

| Algorithm | Split | $n$ | Accept rate | Median (m) | MAE (m) | ≤ 1.5 m | Coverage |
|---|---|---|---|---|---|---|---|
| Trapezoid | Train | 274 | 57% | 0.18 | 0.39 | 97.4% | 90.1% |
| Trapezoid | Test | 91 | 84% | 0.22 | 0.38 | 97.8% | 96.7% |
| ZUPT | Train | 431 | 90% | 0.25 | 0.78 | 90.5% | 92.8% |
| ZUPT | Test | 101 | 94% | 0.17 | 0.47 | 94.1% | 96.0% |

Table 3. Deployed pipeline on accepted predictions only (segmenter → predictor → quality filter). The filter trims the accepted-ride mean error — to 0.38 m for the trapezoid fit and 0.47 m for ZUPT on the held-out test — lifts the one-floor-target rate to 94–98 % there, and holds realised coverage at or above the nominal 90 % on every row. Every false positive on the held-out test (9 of 108 segmenter predictions) is rejected by the filter; the consumer sees zero phantom rides on `Beit Yitzchaki B`.

The accepted-only error CDF and FP diagnostics are in Appendix E.5.

## 6 Discussion

The trapezoid family generalises out of distribution. On the held-out building `Beit Yitzchaki B` the pipeline reaches segmentation $F1^* = 0.88$, a per-ride median trapezoid error of 0.22 m, and zero false positives after the quality filter (Tables 1, 3). Every test-split score matches or exceeds its train counterpart, and correcting for the ±1 s ground-truth pad raises the raw IoU-F1@0.5 from 0.83 to roughly 0.95 (Figure 10). Realised coverage reaches 96.7 % at a nominal 90 %, and the quality filter trims the accepted-ride trapezoid mean error to 0.38 m on the test split (Tables 2, 3). The reason is physical: $j_{\max}$, $a_{\max}$, and $v_{\max}$ come from elevator design standards, so a family built on those limits applies to any compliant cabin.

Three limits bound the scope of this result. First, the held-out evaluation rests on one building — `Beit Yitzchaki B`, a residential block — and held-out office and hotel splits remain future work. Second, the pipeline returns signed displacement $\Delta h$, not a floor index; mapping a sequence of $\Delta h$ to specific floors needs a downstream building-height model the user supplies. Third, the family assumes ISO 18738-compliant kinematics; hydraulic lifts, older geared machines, and freight cabins violate the $(j_{\max}, a_{\max}, v_{\max})$ bounds of (1), and the quality filter rejects those rides at the gates (§4.2.5).

Realised coverage runs 92–98 % on every row of Table 3, above the nominal 90 %, so the conformal multipliers ($k = 5.04$ for trapezoid, $k = 2.34$ for ZUPT) are conservative. They will tighten on a larger calibration set: we fit the trapezoid $k$ on 251 accepted train rides and the ZUPT $k$ on 332, and split-conformal intervals shrink as $1/\sqrt{n_{\text{cal}}}$ (§4.2.4). Tighter coverage trades against recall: the same filter rejects 18 % of clean trapezoid rides and 7 % of clean ZUPT rides on the test split (Table 3 accept rates 82 % and 93 %). Each gate threshold in Tables 6, 7, and 8 is a single scalar, so a deployment can pick its own coverage–recall operating point.

## 7 Conclusion

We turned a smartphone into a per-ride elevator-displacement sensor with a calibrated 90 % confidence interval, accelerometer-only, on a held-out residential building. The mechanical limits of the cabin force the gravity-removed acceleration into a small parametric family, and segmentation and prediction are two fits against that family. Median trapezoid error on the blind test sits at 0.22 m, 96.6 % of accepted rides land within one floor, and the quality filter passes zero phantom rides through to the consumer (Table 3). We release the pipeline, the labelled multi-building dataset, and the evaluation harness — **[TODO: repository URL]**.

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## A Theoretical Elevator Pulse Analysis

This appendix derives the constants of §2.4 from the bounds in (1). §A.1 shows the optimal jerk is bang-bang; §A.2 integrates the resulting seven-segment profile to obtain $T_j$, $T_a$, $T_v$, $T_{\text{ride}}$, and the centroid-spacing identity used in §2.5.

### A.1 The optimal elevator pulse

The time-optimal control problem (2) forces the optimal jerk to take only three values, $j^*(t) \in \{-j_{\max}, 0, +j_{\max}\}$ (3). We sketch the three moves Pontryagin's maximum principle turns on; the full machinery is standard for time-optimal control of a chain of integrators with bounded input [1, 7].

*Hamiltonian.* Stack the state $x = (h, v, a)^\top$ with dynamics $\dot{x} = (v,\ a,\ j)^\top$, control $j$, free terminal time $T$, and unit running cost. Adjoining co-states $\lambda = (\lambda_h, \lambda_v, \lambda_a)^\top$ gives

$$\mathcal{H}(x, \lambda, j) \ = \ 1 + \lambda_h v + \lambda_v a + \lambda_a j,$$

which is linear in $j$.

*Switching law.* Linearity in $j$ pins the PMP-minimising control to the boundary $|j| = j_{\max}$ whenever $\lambda_a \neq 0$:

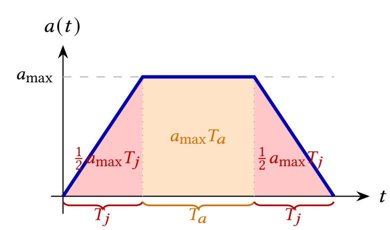
$$j^*(t) \ = \ -j_{\max}\ \operatorname{sign} \lambda_a(t).$$

The co-state equations $\dot{\lambda} = -\partial\mathcal{H}/\partial x$ yield $\lambda_a(t) = \frac{1}{2}\lambda_h t^2 - \lambda_v(0)\, t + \lambda_a(0)$, a quadratic that crosses zero at most twice; the unconstrained $j^*$ therefore switches sign at most twice.

*Constrained arcs.* When $|a| = a_{\max}$ activates, the cabin rides the boundary and the controller must hold $\dot{a} = j = 0$; the velocity constraint $|v| = v_{\max}$ similarly forces $a = 0$ and hence $j = 0$. Stitching the unconstrained $\pm j_{\max}$ arcs with these constrained zero-jerk arcs yields (3) and the three regimes selected by the floor-to-floor height $H$ (Figure 2).

### A.2 Time parameters formula development

We integrate the bang-bang jerk over the take-off lobe (phases 1–3), recover the three durations, and sum to the total ride time. We assume the trapezoidal regime $T_v > 0$; the medium and short regimes are limits $H \downarrow a_{\max} T_j (T_j + T_a)$ of the same formulas. Each phase paints one slab of the lobe beside: the two red wedges of phases 1 and 3, the orange plateau of phase 2. Their areas are the velocity gained on that phase, and the colours below match the equations.




*Phase 1 — jerk ramp ($j = +j_{\max}$).* With $a(0) = v(0) = 0$, one integration gives

$$a(t) = j_{\max}\, t, \qquad a(T_j) = a_{\max} \implies T_j = \frac{a_{\max}}{j_{\max}} \quad (4).$$

Integrating twice more,

$$v(T_j) = \tfrac{1}{2}\, a_{\max} T_j, \qquad h(T_j) = \tfrac{1}{6}\, a_{\max} T_j^2.$$

The velocity $v(T_j)$ is the area of the phase-1 red wedge.

*Phase 2 — acceleration plateau ($a = a_{\max}$).* On $t \in [T_j, T_j + T_a]$ the cabin holds at $a_{\max}$, so the velocity gains the area of the orange rectangle,

$$v(T_j + T_a) \;=\; v(T_j) + a_{\max} T_a \;=\; a_{\max}\left(\tfrac{1}{2} T_j + T_a\right),$$

and the displacement added is $\Delta h_2 = v(T_j)\, T_a + \frac{1}{2} a_{\max} T_a^2$.

*Phase 3 — jerk ramp-down ($j = -j_{\max}$).* On $t \in [T_j + T_a, 2T_j + T_a]$, $a(t) = a_{\max} - j_{\max}(t - T_j - T_a)$ so $a(2T_j + T_a) = 0$ as required. Phase 3 contributes a second $\frac{1}{2} a_{\max} T_j$ of velocity — the mirror red wedge — so the lobe-end velocity is the sum of three slab areas,

$$v(2T_j + T_a) \;=\; \tfrac{1}{2} a_{\max} T_j + a_{\max} T_a + \tfrac{1}{2} a_{\max} T_j \;=\; a_{\max}(T_j + T_a).$$

Demanding the cruise enter at the velocity ceiling, $a_{\max}(T_j + T_a) = v_{\max}$, gives $T_a = \frac{v_{\max}}{a_{\max}} - T_j$ (5). The displacement added in phase 3 is $\Delta h_3 = v(T_j + T_a)\, T_j + \frac{1}{2} a_{\max} T_j^2 - \frac{1}{6} j_{\max} T_j^3$.

*Synthesis.* Summing phases 1–3 with $j_{\max} T_j = a_{\max}$ and $a_{\max}(T_j + T_a) = v_{\max}$ collapses to

$$d_{\text{accel}} \;=\; h(T_j) + \Delta h_2 + \Delta h_3 \;=\; \tfrac{1}{2}\, v_{\max}\,(2T_j + T_a).$$

By symmetry phases 5–7 contribute the same $d_{\text{accel}}$. The cruise covers the remainder $H - 2d_{\text{accel}}$ at $v_{\max}$,

$$T_v \;=\; \frac{H - v_{\max}(2T_j + T_a)}{v_{\max}} \;=\; \frac{H}{v_{\max}} - 2T_j - T_a \quad (6),$$

and $T_{\text{ride}} = 4T_j + 2T_a + T_v$ collapses to the additive form (7).

*Centroid spacing of the two acceleration lobes.* The first lobe spans $[0,\, 2T_j + T_a]$ with centroid at $t_{c,1} = T_j + T_a/2$ (trapezoid symmetry); the second spans $[2T_j + T_a + T_v,\; 4T_j + 2T_a + T_v]$ with centroid at $t_{c,2} = 3T_j + 3T_a/2 + T_v$. Subtracting and using (6),

$$\Delta t_c \;=\; t_{c,2} - t_{c,1} \;=\; 2T_j + T_a + T_v \;=\; \frac{H}{v_{\max}}.$$

The temporal gap between the two acceleration lobes equals $H/v_{\max}$ and is independent of $j_{\max}$ and $a_{\max}$ — the amplitude-free centroid identity used in the basis-function mapping of §2.5.

### A.3 Template-grid bounds from literature machine parameters

The segmentation appendix's template grid (Appendix C.1) ranges over a discrete set of $(W, f)$ cells. The grid limits are not free parameters — they fall out of the kinematic identities (10)–(11) once the literature ranges for the three machine parameters are plugged in.

*Literature ranges.* Section 2.2 cites the following bounds for commercial passenger elevators:

$$j_{\max} \in [1.0,\ 1.6]\ \text{m/s}^3\ [6], \qquad a_{\max} \in [0.8,\ 1.5]\ \text{m/s}^2, \qquad v_{\max} \in [1.0,\ 2.5]\ \text{m/s}\ [4]. \tag{33}$$

*Time constants.* Equation (4) gives the jerk ramp $T_j = a_{\max}/j_{\max}$, whose range under (33) is

$$T_j \;\in\; \left[\frac{a_{\min}}{j_{\max}},\ \frac{a_{\max}}{j_{\min}}\right] \;=\; \left[0.50,\ 1.50\right]\ \text{s}.$$

Equation (5) gives the acceleration plateau $T_a = \max(0,\, v_{\max}/a_{\max} - T_j)$; its lower bound is 0 (any configuration with $v_{\max}/a_{\max} \le T_j$ saturates the $a_{\max}$ rail but never reaches $v_{\max}$), and its upper bound is attained at $v_{\max}=2.5$, $a_{\max}=0.8$, $j_{\max}=1.6$, yielding

$$T_a \;\in\; \big[0,\; 2.63\big]\ \mathrm{s}.$$

*Half-width $W$ and flat fraction $f$.* Substituting the eight corner combinations of (33) into (10)–(11) yields

$$W \;\in\; \big[0.88,\; 1.96\big]\ \mathrm{s}, \qquad f \;\in\; \big[0,\; 0.72\big].$$

The lower end of $f$ is the triangle limit ($T_a = 0$, short hops that never reach the $a_{\max}$ plateau); the upper end is the near-rectangular plateau of the longest cruises.

*The deployed grid.* The detector pads these analytical bounds to absorb sensor noise, controllers that depart from the textbook bang-bang law, and a margin for the short-ride triangle row. The deployed grid is 30 log-spaced $W$ in $[0.4,\ 3.0]$ s and 15 linear $f$ in $[0.05,\ 0.80]$, plus an explicit $f=0$ triangle row for short rides that never reach the $a_{\max}$ plateau. The same grid is reused by every stage that fits the trapezoid family (Appendices C.1 and D).

## B Dataset collection and ground-truth pipeline

The dataset is 195 labelled elevator rides drawn from eight recording sessions across seven elevators in six buildings, all collected in March–April 2026. This appendix walks the construction pipeline end-to-end: how each session was recorded and how every ride got its label and floor-to-floor $\Delta h$ (§B.1), the human-in-the-loop edit pass that corrects detector mistakes (§B.2), and the per-session statistics of the campaign (§B.3).

### B.1 Method

*Recording setup.* Each session ran four phones simultaneously, carried together in one bag or hand so every device saw the same motion, vibration, and acoustics. Two experimenters carried two phones each, and the fleet always included at least one barometer-equipped device whose pressure trace became the ground truth for the three accelerometer-only phones. Two of the eight sessions — Haari 3 and Bar-Ilan 2 — relied on a single experimenter holding one barometer-equipped phone. Table 4 lists the channels each phone logged at its native rate; the loader resamples every channel to 50 Hz before downstream stages see it.

*Barometric pressure to altitude.* The recording phone's pressure channel converts to a metric altitude trace through the International Standard Atmosphere troposphere inversion, $h = (T_0/L)\,[\,1 - (P/P_0)^{1/5.255}\,]$, with the surface temperature $T_0$ read from the session log. Figure 12 plots both sides on a Millennium-Hotel session. Pressure climbs by ~6 hPa as the experimenter descends from floor 12 to the ground lobby, and the inverted altitude resolves the matching ~50 m descent on the same time axis. At standard $T_0$ the resolution sits near 10 cm per sample, an order of magnitude below the ~3 m typical inter-floor pitch, so floor-to-floor steps stand clear of sensor noise.

*Edge-point ride extraction.* `HeightSegmenter` turns the altitude trace into ride intervals through a deterministic state machine over vertical velocity. The procedure low-pass filters altitude across an 8 s rolling window, differences the result into velocity, smooths velocity with a 3 s box kernel, and marks samples above $|v_z| > 0.15$ m/s as moving. Same-sign moving runs within 6 s of each other merge into one interval. The detector then drops any survivor shorter than 3 s or with $|\Delta z| < 2$ m. The pipeline pads each remaining survivor by 1 s on either side and labels it `up` when the mean velocity is positive, `down` otherwise; gaps carry the `outside` label, so every sample wears one of the three tags. Figure 13 runs

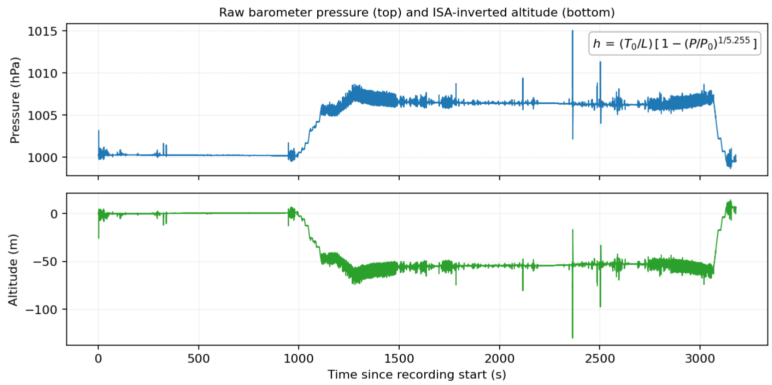



Fig. 12. Pressure (**top**) and the ISA-inverted altitude (**bottom**) on a Millennium-Hotel session. The pressure rise around $t \approx 1000$ s is the experimenter taking the elevator down from floor 12; the inverted altitude shows the matching ~50 m descent.

the procedure on a 170 s window of Millennium-Hotel session exp2. Four step-shaped descents in the altitude panel become four negative-velocity dips that breach the threshold, and the bottom panel emits the matching down intervals interleaved with `outside` gaps. Every operation is deterministic, so repeated runs label a recording identically.

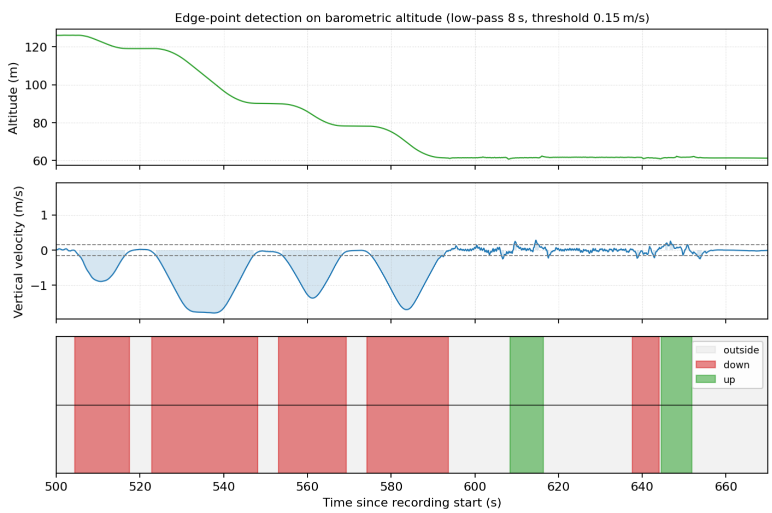



Fig. 13. Barometer interval detector on a 170 s window of Millennium-Hotel session exp2. **Top:** low-pass altitude. **Middle:** smoothed vertical velocity with the ±0.15 m/s threshold (dashed) and the "moving" regions shaded. **Bottom:** the surviving up/down intervals after merging, with `outside` between them.

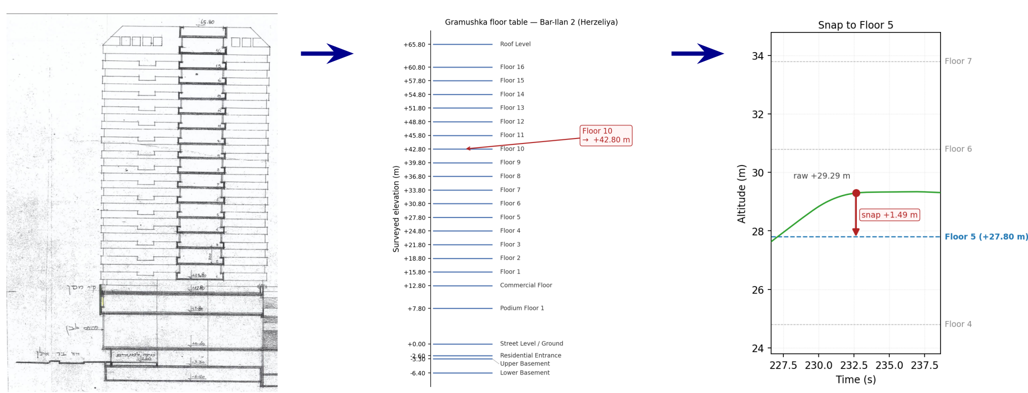
*Sensor channels.* An Android logger subscribes to every inertial and environmental channel the phone exposes and writes one tab-separated row per sample (Table 4). The pipeline reads accelerometer (ACC) on every phone, barometer (PRS) on the equipped phone for ground truth, and gyroscope, magnetometer, and orientation for the editor's secondary plots and the cross-device alignment step.

| Tag | Sensor | Columns |
|---|---|---|
| `ACC` | Linear accelerometer | `x, y, z` |
| `GYR` | Gyroscope | `x, y, z` |
| `MAG` | Magnetometer | `x, y, z` |
| `ORI` | Rotation-vector quaternion | `w, x, y, z` |
| `PRS` | Barometer | `pressure` |
| `GPS` | GPS fix | `lat, lon, alt` |

Table 4. Sensor channels emitted by the logger. The pipeline reads `ACC` throughout; `PRS` supplies ground truth only.

*Cross-device time synchronisation.* Android stamps each sample against device boot time, not wall-clock, so two phones in the same session disagree on event time by their boot-difference — often tens of thousands of seconds. The loader anchors each log on its first-sample timestamp to remove the gross offset, then sharpens the alignment with a normalised cross-correlation of the two accelerometer magnitudes across a ±10 s window. After correction the residual lag sits below one 50 Hz sample, and the pipeline bakes the recovered offset into the secondary barometer stream before merging.

*Gramushka tables and floor snapping.* Raw barometer altitude drifts with weather and building airflow, so the pipeline snaps every ride endpoint to a building-specific floor table. Figure 14 threads the three steps. Each building contributes a *gramushka* — the team's nickname for the accordion-folded architectural section drawing (Figure 14a). Transcribing the drawing's labelled floor lines yields a lookup from floor name to surveyed elevation (Figure 14b). For each detected ride the pipeline integrates the temperature-aware altitude across the interval, snaps the endpoint to the nearest floor in the lookup, and treats the resulting floor-to-floor displacement as the ground-truth $\Delta h$ (Figure 14c; the worked example reads raw +29.29 m and snaps +1.49 m down to Floor 5 at +27.80 m). The external Millennium elevator has no drawing; its rides keep the drift-corrected barometer $\Delta h$ directly.




(a) Architectural drawing (b) Floor-height lookup (c) Snap to nearest floor

Fig. 14. From drawing to floor-height lookup to snapped ride endpoint. (a) The accordion-folded architectural section drawing for one campaign building (Bar-Ilan 2, Herzeliya). (b) The transcribed lookup, mapping each floor name to one surveyed elevation (annotated arrow: Floor 10 at +42.80 m). (c) A worked snap from a real Bar-Ilan 2 ride: the raw barometer altitude +29.29 m snaps +1.49 m down to Floor 5 at +27.80 m.

## B.2 Hand editing

*Why hand editing.* The barometer detector cannot tell elevator motion from any other vertical motion of the same magnitude. Stair climbs, driving up or down a hill, and HVAC pressure ripples each pass the velocity threshold and slip into the label stream; conversely, an over-tight merge fuses two close rides into one. Experimenters carry side information — which button they pressed, which lobby they entered, which staircase they took — that no sensor sees, and the hand-edit pass injects that knowledge into the labels.

*The editor.* Every recording passes through the ground-truth editor of Figure 15 at least once. Stacked panels show altitude, vertical velocity, and the accelerometer, gyroscope, and magnetometer magnitudes on a shared time axis; ride intervals overlay as coloured spans (green = up, red = down, grey = outside). A side table lists every interval for direct editing, and the detector's current proposals appear as toggleable hatched bands the editor can promote, redraw, or reject.

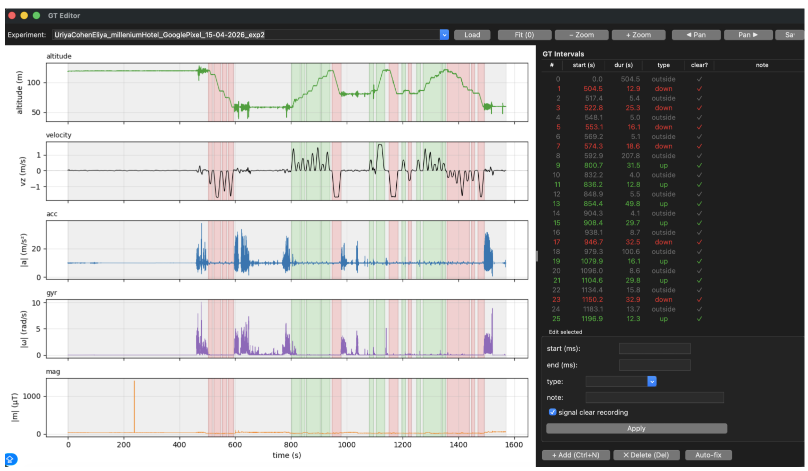



Fig. 15. The ground-truth editor. Stacked panels show altitude, vertical velocity, and the accelerometer, gyroscope, and magnetometer magnitudes on a shared time axis; ground-truth intervals are overlaid as coloured spans (green = up, red = down, grey = outside). The side table lists every interval for direct editing.

*What the editor fixes.* The pass deletes stair-climb and driving-induced intervals the detector mistook for rides, splits intervals where two consecutive rides merged across a short pause, and retags any direction the velocity-sign rule flipped from a low-amplitude pre-ride drift. Dragging a mis-aligned endpoint on the plot snaps it onto the altitude step; the editor commits the GT CSV on save. Every ground-truth file in the dataset has at least one human-confirmed pass.

## B.3 Buildings and phones

The campaign covers seven elevators across six buildings in three Israeli cities — Ra'anana, Herzeliya, and Ramat Gan — yielding 195 labelled rides across eight sessions. Six of the eight sessions ran the standard four-phone fleet (two experimenters × two phones each): Google Pixel 10, Samsung Galaxy A23, Samsung Galaxy S23, and Xiaomi Redmi Note 12. The Haari 3 and Bar-Ilan 2 sessions deviated to a single barometer-equipped phone (Samsung Galaxy Z Flip6 and Google Pixel 10 respectively). Ride counts per session range from 10 (Millennium Hotel exp1) to 44 (Bar-Ilan 2);

Table 5 lists every session with its phones, latitude and longitude where surveyed, building type, and ride count. `Beit Yitzchaki B` is the held-out blind-test building; the other seven sessions feed training.

| Session | Type | Lat, Long | Phones | Rides |
|---|---|---|---|---|
| Haari 3, Ramat Gan, 2026-04-10 | Residential | 32.0658, 34.8233 | Z Flip6 | 39 |
| Bar-Ilan 2, Herzeliya, 2026-03-24 | Residential | 32.166394, 34.842190 | Pixel 10 | 44 |
| Millennium Hotel, exp1, 2026-04-15 | Hotel | — | Pixel 10, Note 12, A23, S23 | 10 |
| Millennium Hotel, exp2, 2026-04-15 | Hotel | — | Pixel 10, A23, S23, Note 12 | 31 |
| Millennium Outside, exp3, 2026-04-15 | Hotel (external) | — | Pixel 10, A23, S23, Note 12 | 12 |
| Acro Building, exp4, 2026-04-15 | Office | — | Pixel 10, A23, S23, Note 12 | 12 |
| Beit Mansour 1, exp5, 2026-04-15 | Residential | — | Pixel 10, A23, S23, Note 12 | 18 |
| Beit Yitzchaki B, exp6, 2026-04-15 | Residential (test) | — | Pixel 10, A23, S23, Note 12 | 29 |

Table 5. The eight recording sessions of the field campaign. Phone short names: Pixel 10 = Google Pixel 10, A23 = Samsung Galaxy A23 (SM-A235F), S23 = Samsung Galaxy S23 (SM-S911B), Note 12 = Xiaomi Redmi Note 12 (22101320I), Z Flip6 = Samsung Galaxy Z Flip6. Latitude and longitude are recorded for two buildings; the remaining four are kept anonymised pending publication.

A companion data drop at **[TODO: <DRIVE-URL>]** mirrors the complete raw sensor logs, gramushka tables, per-session ground-truth CSVs, and editor screenshots for reproducibility.

## C Segmentation: matched-filter and pair-filter details

This appendix derives the matched-filter solution and the shared-shape joint score of §4.1, then walks every calibrated threshold of Table 6 row by row. Each subsection takes one row of the table, explains the intuition behind the test, and — where the picture is worth more than the prose — carries a single illustrative figure.

| **Stage / gate** | **Condition** | **Threshold** |
|---|---|---|
| Detect: shape floor | $R^2(i) \geq$ | 0.40 |
| Detect: amplitude floor | $\lvert\hat{A}(i)\rvert \geq$ | $0.20\,\mathrm{m/s^2}$ |
| Detect: peak dedup | non-max suppression radius | 1.0 s |
| Detect: same-sign gap | min. gap, equal signs | 5.0 s |
| Gate 1: temporal gap | $t(i_2) - t(i_1) \in$ | $[0, 30]$ s |
| Gate 2: joint score | $\frac{1}{2}(R_1^2 + R_2^2) \geq$ | 0.90 |
| Gate 3: shared amplitude | $A^\star \geq$ | $0.30\,\mathrm{m/s^2}$ |
| Gate 4: grid energy | $E(i, j) \geq$ | 0.40 |
| Gate 5: quiet middle | $\mathrm{RMS_{mid}}/A^\star \leq$ | 0.5 |
| Gate 6: duration penalty | rank $= S - \lambda\,\Delta t$ | $\lambda = 0.01\,\mathrm{s^{-1}}$ |

Table 6. Calibrated detection and pair-filter thresholds. The kinematics fix the template-grid bounds; a grid search on the training experiments chose the acceptance thresholds against a composite segmentation metric. The values shown are the detector's deployed defaults.

### C.1 Matched-filter solution and template grid

Before the row-by-row walk, we collect the three pieces of machinery the gates all rely on: the per-window least-squares fit, the $(W, f)$ template grid that fit ranges over, and the shared-shape joint score that turns two candidate lobes into one pair fit.

*Matched-filter least-squares solution.* The detector fits each window $a_{[i]}$ with a single scaled template, minimising $\|a_{[i]} - A\,\tau_{W,f}\|^2$ over $A$. The normal equation $A\,\|\tau_{W,f}\|^2 = \langle a_{[i]}, \tau_{W,f}\rangle$ gives the fitted amplitude and the coefficient of determination of (17). The score $R^2_{W,f}(i)$ is the window-energy fraction the template explains: $R^2 = 1$ an exact fit, $R^2 \leq 0$ no better than the zero signal.

*Template grid.* Appendix A.3 derives the analytical $(W, f)$ ranges admitted by the commercial machine parameters of §2.2; the deployed grid pads them to 30 log-spaced $W$ in $[0.4,\ 3.0]$ s and 15 linear $f$ in $[0.05,\ 0.80]$, plus an $f = 0$ triangle row for short rides that never reach the $a_{\max}$ plateau.

*Shared-shape joint score.* The pair score (21) fits *one* amplitude to *both* lobes of a candidate pair. With $a_k$ the $k$th lobe window, $P_k = \|a_k\|^2$, $u_k = s_k\langle a_k, \tau_{W,f}\rangle$ the sign-corrected correlation, and $N = \|\tau_{W,f}\|^2$, minimising $\sum_k \|a_k - s_k A\,\tau_{W,f}\|^2$ over the shared $A$ gives $A^\star = (u_1 + u_2)/(2N)$ and the per-lobe $R_k^2$ of (21). Unrelated bursts peak at a different cell for each lobe, so forcing one $(W, f)$ on both is what makes the score discriminating.

### C.2 Detection shape and amplitude floors

The first two rows of Table 6 reject samples that cannot be a take-off or landing lobe at all (19). The shape floor $r_{\min} = 0.40$ is deliberately permissive: hand-held jitter and clipped windows pull a real lobe's $R^2$ below a textbook trapezoid's, so the detector leaves the strict shape test to the pair stage. The amplitude floor $a_{\min} = 0.20\,\mathrm{m/s^2}$ drops lobes buried in sensor and walking noise, and rises per phone when a device's measured noise runs higher. Both favour recall — a missed lobe is lost for good, while a spurious one still faces the pairing gates.

### C.3 Peak de-duplication and same-sign gap

The next two rows of Table 6 stop one physical lobe from entering the pair stage as several peaks. A lobe's matched-filter score forms a ridge several samples wide, so non-maximum suppression over a 1.0 s radius keeps only its crest, and the same-sign gap collapses any two *equal*-sign peaks closer than 5.0 s — a lobe re-detected at a neighbouring $W$. Opposite-sign peaks pass untouched: separating a take-off from its landing is the pairing stage's task.

### C.4 Gate 1 — pairing window $[t_{\min}, t_{\max}]$

Gate 1 bounds how far a take-off and a landing may sit apart before the pair is scored (20) — a search window, not a model of ride duration. The lower bound $t_{\min} = 0$ s admits the shortest single-floor hops, whose lobes almost touch; the upper bound $t_{\max} = 30$ s brackets one uninterrupted ride at the car speeds and inter-floor heights of vertical-transportation practice [4, 13]. A high-rise express run would need a wider window; the 30 s cap trades that reach against joining lobes drawn from *different* rides.

### C.5 Gates 2 & 3 — joint shape and shared amplitude

Gates 2 and 3 read the same shared-shape fit (21) from two different angles: a single $(W, f)$ template explaining the *shapes* of both lobes, and the mean $A^\star$ summing their *amplitudes*. Both lobes are cut from one cabin profile, so a real ride must clear both gates; an accidental pairing typically fails at least one.

Gate 2 demands a joint score of at least $r_{\text{pair}} = 0.90$, far above the 0.40 detector floor. Unrelated bursts each fit some template alone but peak at different cells, so no shared $(W, f)$ scores both above 0.90.

Gate 3 requires the shared amplitude $A^\star$ to clear $a_{\text{pair}} = 0.30\,\mathrm{m/s^2}$, above the 0.20 detector floor. Since $A^\star$ is the mean of the two lobe amplitudes, a pair survives only when *both* lobes are firm: a strong take-off matched to a weak noise lobe averages below 0.30, even when the joint shape still passes Gate 2.

Together the two floors are cheap — they cost almost no true rides while removing the bulk of accidental pairings (Figure 16).

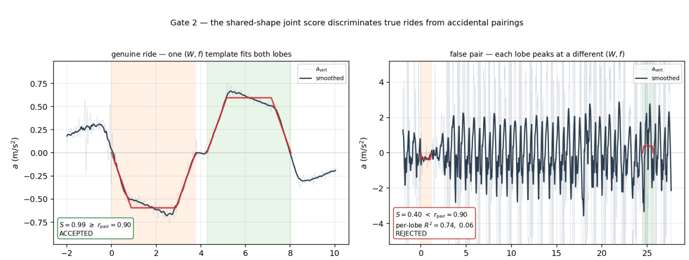



Fig. 16. Gates 2 and 3 on two real candidate pairs. **Left:** a genuine ride — one $(W, f)$ trapezoid (red) overlays both opposite-sign lobes; $S = 0.99$ and $A^\star$ both clear their floors. **Right:** an accidental pairing — one $(W, f)$ fits one lobe well ($R_1^2 = 0.74$) but not the other ($R_2^2 = 0.06$); $S = 0.40$ falls below the joint-score floor and the weak lobe drags $A^\star$ below its floor too.

### C.6 Gate 4 — grid energy $e_{\min}$

Gate 4 measures *how broadly* the grid supports the match. A genuine ride near-fits a whole neighbourhood of $(W, f)$ templates — a wide bright band in the joint-$R^2$ heatmap — while a coincidental pairing fits one lucky cell and leaves the grid dark. Grid energy $E$ averages the clamped joint $R^2$ over every cell (23); the floor $e_{\min} = 0.40$ sits above a single-cell match and below any broad band, catching false pairs that clear the best-cell score of Gate 2 (Figure 17).

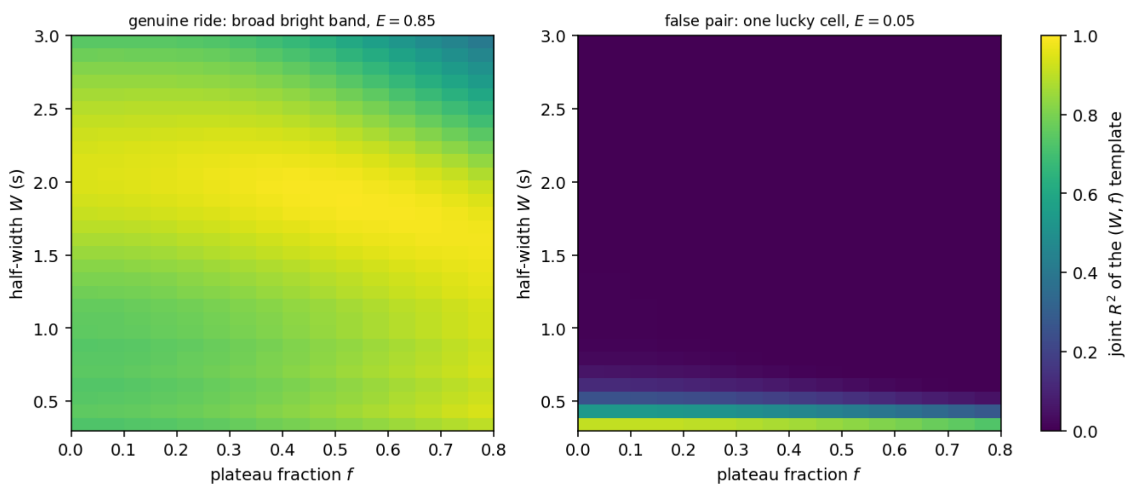



Fig. 17. The grid-energy gate $e_{\min}$. Each panel is the joint-$R^2$ heatmap over the $(W, f)$ template grid for one candidate pair. A genuine ride (left) is a near-fit across a broad band of templates, so its grid energy $E$ sits well above the 0.40 floor; a coincidental pairing (right) fits one narrow sliver and leaves the grid dark, so its energy falls far below the floor and Gate 4 rejects it.

### C.7 Gate 5 — quiet-middle cruise gate $\rho$

Between its two lobes a real ride holds constant velocity, so the vertical acceleration there is near zero. Gate 5 takes $\mathrm{RMS}_{\mathrm{mid}}$, the root-mean-square of the smoothed acceleration on the inter-lobe window $(t_{c_1} + W,\ t_{c_2} - W)$, and rejects the pair when it exceeds $\rho A^{\star}$ with $\rho = 0.5$ — when the cruise wobble passes half the lobe amplitude. Equivalently, fit a line through the cruise samples; the RMS rule rejects pairs whose regression tilts by more than $\approx 15^{\circ}$ from horizontal (Figure 18). The test catches pairs whose middle never settles, most often a walking false positive where continuous motion keeps the inter-lobe stretch as loud as the lobes.

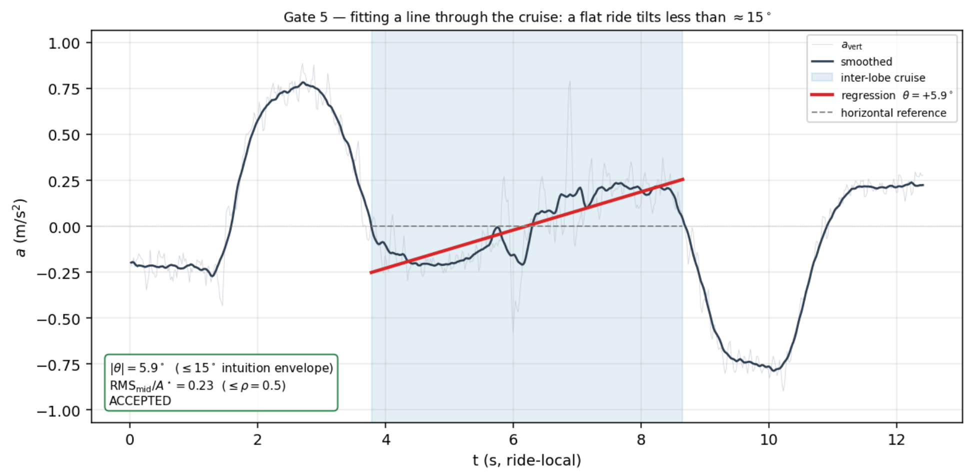



Fig. 18. Gate 5 on the first short ride of a real multi-stop trip (the same recording that drives Figure 19). A least-squares line through the inter-lobe cruise samples (shaded) tilts by only $\theta = 5.9°$, well inside the $\approx 15°$ envelope; equivalently $\mathrm{RMS}_{\mathrm{mid}}/A^{\star} = 0.23$ is well under $\rho = 0.5$, and the pair is accepted.

## C.8 Gate 6 — duration penalty $\lambda$ and greedy resolution

A passenger often rides in stages — floor 1 to 4, a stop, then 4 to 8 — recorded as two back-to-back rides and *four* lobes (Figure 19). The take-off of the first ride and the landing of the last form a *super pair*: it shares the same cabin shape, so its joint score $S$ matches either true ride and Gate 2 cannot tell them apart. Gate 6 breaks the tie by ranking every admissible pair by $S - \lambda\,\Delta t$ with $\lambda = 0.01\,\mathrm{s}^{-1}$ (24): the penalty docks the long span far more than the short ones, so the two short rides out-rank the super pair. A greedy resolver commits pairs highest-first, skipping any that reuse a committed lobe, and the super pair never commits.

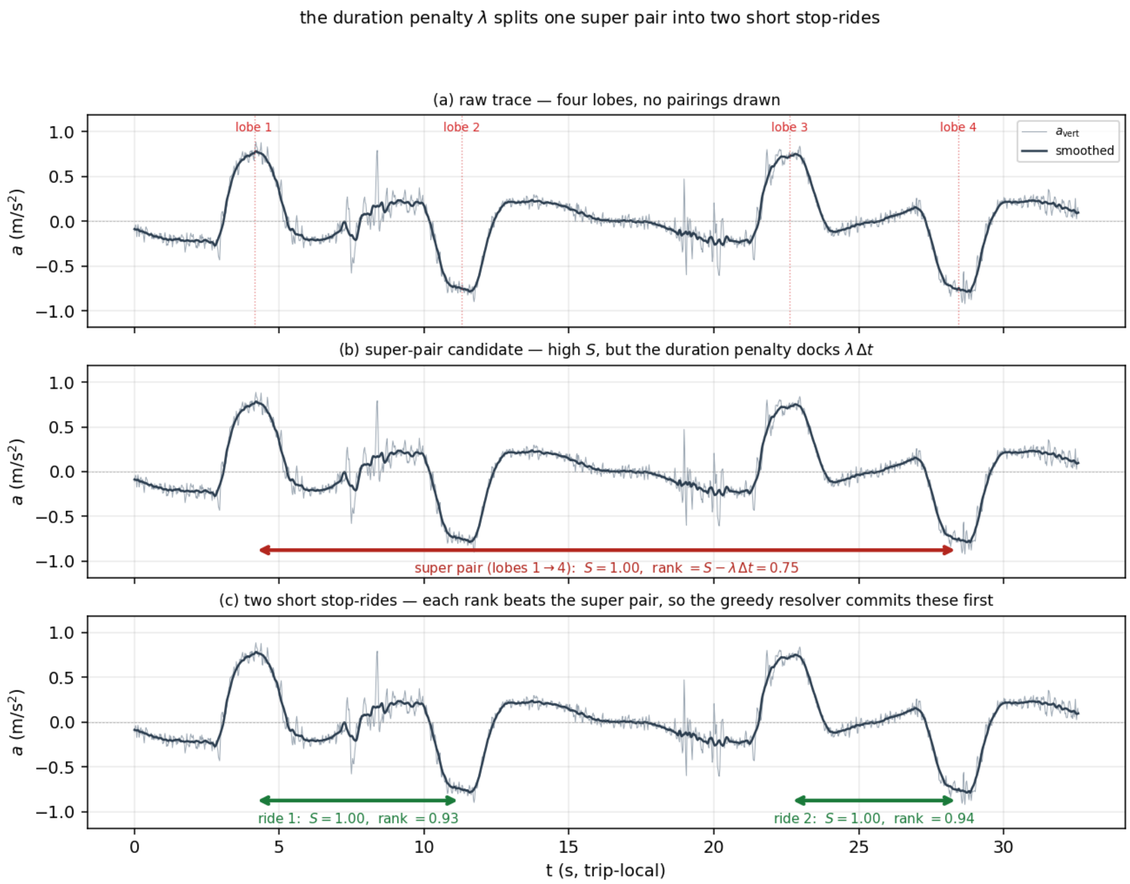



Fig. 19. The duration penalty $\lambda$ on an up-going multi-stop trip, reorganised into three panels. **(a)** The bare acceleration trace: four lobes, no pairings drawn. **(b)** The *super-pair* candidate (red arrow): the take-off of the first ride paired with the landing of the last. Its joint score $S$ is high, but the rank $S - \lambda\,\Delta t$ is docked by the wide span. **(c)** The two short stop-rides (green arrows): each carries a comparable $S$ but a far smaller $\Delta t$, so each rank beats the super pair. The greedy resolver commits the two short rides first, and the super pair never makes it onto the page of detected intervals.

## D Prediction: error budget and conformal calibration

This appendix derives the closed-form height (27), the per-algorithm error budgets of §4.2.3, and the conformal coverage guarantee of (32), and it explains and illustrates every quality-filter check of §4.2.5.

### D.1 The trapezoid pulse-pair integral

The closed-form height (27) reduces to one elementary scalar — the trapezoid area — used three times: once per lobe and once for the cruise.

*Lobe area.* The trapezoid $\tau_{W,f}$ of (9) vanishes outside $[-W, W]$ and is a plateau of width $2fW$ joined to zero by two ramps of base $(1-f)W$, so

$$\int_{-\infty}^{\infty} \tau_{W,f}(t)\, dt \;=\; \underbrace{2fW}_{\text{plateau}} \;+\; \underbrace{2 \cdot \tfrac{1}{2}(1-f)W}_{\text{two ramps}} \;=\; W(1+f). \tag{34}$$

One lobe $sA\,\tau_{W,f}$ therefore steps the cabin velocity by $v_{\text{peak}} = A\,W(1+f)$: the take-off lobe carries the cabin from rest to $+v_{\text{peak}}$, the landing lobe back to rest.

*Displacement.* The velocity $v(t)$ rises through the take-off lobe, holds at $v_{\text{peak}}$ across the inter-lobe cruise of duration $T_v = \Delta t_c - 2W$, then falls through the landing lobe (Figure 20). The cruise contributes $v_{\text{peak}}\,T_v$ to $\Delta h$. Each ramp window contributes $v_{\text{peak}}\,W$: $\tau_{W,f}$ is symmetric about its centre, so $v(t)$ is point-symmetric about each centroid with $v(t_{c,k}) = v_{\text{peak}}/2$, making its mean across the $2W$-wide lobe window exactly $v_{\text{peak}}/2$. Summing the three slabs,

$$\Delta h \;=\; s\,v_{\text{peak}}\,(2W + T_v) \;=\; s\,v_{\text{peak}}\,\Delta t_c \;=\; s\,A\,W(1+f)\,\Delta t_c,$$

which is (27): the cruise velocity times the centroid spacing, signed by the ride direction.

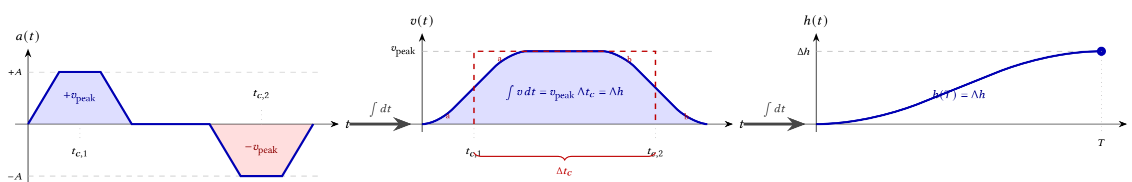



Fig. 20. Double integration of the pulse pair. Each acceleration lobe integrates to a velocity step of $\pm v_{\text{peak}} = \pm AW(1+f)$ centred at $t_{c,1}, t_{c,2}$; $v(t)$ holds at $v_{\text{peak}}$ only on the inter-lobe cruise. Each ramp window is point-symmetric about its centroid, so the shaded sliver outside the dashed rectangle ($a$, $b$) has the same area as the unshaded sliver inside it, making the area beneath $v(t)$ equal to $v_{\text{peak}}\,\Delta t_c = \Delta h$.

In the short-ride limit $\Delta t_c = 2W$ the two lobes touch and the cruise vanishes; (27) collapses to the joined-pulse height

$$\Delta h \;=\; 2\,s\,A\,W^2(1+f), \tag{35}$$

which the predictor fits as a three-parameter alternative on every segment.

## D.2 ZUPT error budget

The ZUPT standard deviation (29) sums four independent variances.

*White-noise double integration.* Integrating a white-noise acceleration sequence of per-sample standard deviation $\sigma_a$ once produces a velocity random walk; integrating again produces a position. After the linear drift term of (25) removes the unconstrained ramp mode, the end-point position variance is $\sigma_a^2\,\Delta t^4\,N^3/12$ over $N$ active samples, whose root is (28). The cube in $N$ is the double-integration random-walk scaling; the factor $1/12$ is the residual after the drift correction, against the $1/3$ of an uncorrected double integral.

*Mechanical jitter.* A hand-held phone is not a rigid inertial platform: hand tremor, pocket friction, and skin-coupled breathing add a coloured-noise acceleration band the datasheet omits. The budget models it as an extra acceleration $\sigma^a_{\text{mech}} = j_0\,(1 + \min(\rho/10°, 2))$, a baseline $j_0 = 0.03\,\text{m/s}^2$ scaled up to threefold by the ride drift angle $\rho$, then integrated under ZUPT exactly as the white-noise term: $\sigma_{\text{mech}} = \sigma^a_{\text{mech}}\,\Delta t^2\sqrt{N^3/12}$.

*Relative scale.* The longer the ride, the larger the absolute error, because the unmodelled coloured drift is roughly linear in the path integral of acceleration. A relative floor $\sigma_{\text{rel}} = k_{\text{rel}}\,|\Delta h|$ with $k_{\text{rel}} = 0.05$ captures it.

*Projection penalty.* A ride that lacks a clean stationary window projects its vertical axis from a one-sided or in-ride gravity estimate, losing one or two degrees of freedom on the vertical direction. The penalty $\sigma_{\text{proj}} = (\pi - 1)\cdot 0.25\,|\Delta h|$ scales with a method factor $\pi \in \{1.0, 1.2, 1.3, 3.0\}$ for the pre-and-post, pre-only, post-only, and magnitude-only projections, further multiplied by $1 + \min(\rho_{\text{pp}}/12.5°, 2)$ for the pre/post rotation angle $\rho_{\text{pp}}$. Combined in quadrature (29) and floored at 0.15 m, the four terms give the per-ride standard deviation

$$\sigma_{\text{ZUPT}} = \max\left(\sqrt{\sigma^2_{\text{white}} + \sigma^2_{\text{mech}} + \sigma^2_{\text{rel}} + \sigma^2_{\text{proj}}},\ 0.15\,\text{m}\right),$$

so the conformal multiplier cannot blow up on a degenerate fit.

### D.3 Trapezoid error budget

*Matched-filter Cramér–Rao bound.* Model the vertical trace as the fitted template pair plus Gaussian noise of standard deviation $\sigma^{\text{eff}}_a$. The Fisher information of this likelihood bounds the variance of each fitted parameter. For the amplitude and the lobe centre,

$$\sigma^2_A = \frac{(\sigma^{\text{eff}}_a)^2}{\langle\tau,\tau\rangle}, \qquad \sigma^2_{t_c} = \frac{(\sigma^{\text{eff}}_a)^2}{A^2\,\langle\dot\tau,\dot\tau\rangle}, \qquad \sigma^2_{\Delta t_c} = 2\,\sigma^2_{t_c}, \tag{36}$$

where $\langle\tau,\tau\rangle$ is the template energy of (17) and $\langle\dot\tau,\dot\tau\rangle = 2/[(1-f)W]$ is the slope energy of the two ramps. A wider, sharper-edged lobe carries more energy and more slope, so it pins both amplitude and timing more tightly. The grid quantises the shape parameters, so their variance is that of a uniform prior over one grid cell, $\sigma^2_W = \Delta^2_W/12$ and $\sigma^2_f = \Delta^2_f/12$.

*Delta-method propagation.* Differentiating the closed-form height (27) gives the four gradients

$$\frac{\partial\Delta h}{\partial A} = sW(1{+}f)\Delta t_c,\quad \frac{\partial\Delta h}{\partial W} = sA(1{+}f)\Delta t_c,\quad \frac{\partial\Delta h}{\partial f} = sAW\Delta t_c,\quad \frac{\partial\Delta h}{\partial\Delta t_c} = sAW(1{+}f), \tag{37}$$

and substituting (36) and (37) into the delta-method sum (30) yields the per-segment variance

$$\sigma^2_{\Delta h} = [W(1{+}f)\,\Delta t_c]^2\,\sigma^2_A + [A(1{+}f)\,\Delta t_c]^2\,\sigma^2_W + [AW\,\Delta t_c]^2\,\sigma^2_f + [AW(1{+}f)]^2\,\sigma^2_{\Delta t_c}.$$

*Data-adaptive effective noise.* The white-noise figure from the sensor datasheet understates the error on a hand-held phone by an order of magnitude. The pipeline replaces it with the residual root-mean-square of the fitted template,

$$\sigma^{\text{eff}}_a = \max\left(\sqrt{\tfrac{1}{N}\,\|a - \hat a_{\theta^\star}\|^2},\ \sigma^{\text{ds}}_a\right),$$

floored at the datasheet value $\sigma^{\text{ds}}_a$. Whatever the template fails to explain — tremor, projection wobble, coloured drift — enters the residual, so the budget counts it automatically. A second correction divides the effective variance by the fit quality, $\tilde\sigma^2_a = (\sigma^{\text{eff}}_a)^2/\max(R^2_{\text{joint}}, \varepsilon)$ with $\varepsilon = 10^{-3}$ — the Wald post-regression form, in which a poorly fitting template widens the posterior on every parameter in proportion.

*Velocity-anchored amplitude variance.* When the fit of §4.2.2 replaces the matched-filter amplitude with the measured cruise velocity, the CRB (36) no longer describes $\sigma_A$. The cruise velocity is a mean of white-noise velocity samples, so $\sigma_{\tilde{v}}^2 = \tilde{\sigma}_a^2\,\Delta t/T_{\text{cruise}}$ and $\sigma_{\tilde{A}}^2 = \sigma_{\tilde{v}}^2/[W(1+f)]^2$, which the pipeline substitutes for $\sigma_A^2$ whenever the cruise window is long enough to anchor ($T_{\text{cruise}} > 0.5$ s).

*Overlap inflation of the centroid spacing.* The CRB (36) treats the two lobe centres as independent. At $\Delta t_c = 2W$ the lobes touch but their supports stay disjoint, the off-diagonal Fisher block is zero, and the diagonal CRB is exact. For $\Delta t_c < 2W$ — unphysical for the shared-shape pair model — the two centre estimates correlate, and the budget inflates the spacing variance by

$$\sigma_{\Delta t_c}^2 = 2\,\sigma_{t_c}^2 \left[ 1 + \left( \frac{2W - \Delta t_c}{2W\,\delta} \right)^2 \right], \qquad \delta = 0.05, \tag{38}$$

and the factor is 1 in the physical regime $\Delta t_c \geq 2W$. The inflation keeps the budget well-conditioned without penalising a genuine touching-lobe ride, which the joined-pulse fit (35) handles directly.

### D.4 Split-conformal coverage and calibration

The multiplier $k$ of §4.2.4 is the $\lceil (1-\alpha)(n+1) \rceil / n$ empirical quantile of the $n$ calibration nonconformity scores (31). For exchangeable rides the rank of a test score among the calibration scores is uniform, so the test score falls at or below that quantile with probability at least $1-\alpha$ — which is (32), with no assumption on the shape of the error distribution.

The two algorithms calibrate $k$ on different pools. ZUPT calibrates on every clean ride: the reported metrics are computed on clean data regardless of the quality verdict, so a consumer who overrides the filter still receives the promised coverage. The trapezoid fit calibrates on the rides that are both clean and quality-accepted — its exchangeable population, since a rejected trapezoid fit carries a $\sigma$ the budget knowingly under-predicts. Exchangeability fails across the clean/corrupted split, so neither pool draws on corrupted recordings. The pipeline floors the multiplier at the Gaussian 90 % half-width 1.645, and keeps an absolute floor and cap on the reported $k\sigma$ as a guard against a pathologically small or large fitted $\sigma$.

### D.5 Quality-filter checks

Each algorithm runs its own quality filter (§4.2.5). The two filters share a shape: hard gates that reject the ride outright at the first breach, and graded score terms that sum into a penalty rejecting the ride above 6. Table 7 lists the ZUPT checks; Table 8 the trapezoid checks. The remainder of this appendix walks each check one at a time: the real-world failure it catches, the quantity it measures, and where the threshold comes from.

*Orientation checks (both filters).* Every accelerometer estimate rides on a single vertical axis. Three checks defend that axis at three different time-scales: before the ride, across the ride, and inside the ride.

*Pre-window stationarity ($s_{\text{pre}}$).* The pipeline averages accelerometer samples in a rest window just before the ride to recover the gravity direction. A still phone holds three flat axis traces; a swinging or rotating phone writes wobble into all three, and the averaged vector then points wherever the wobble happened to drift. The filter measures the spread of those gravity samples. A window the filter cannot certify forces the unsigned-magnitude fallback, and the filter rejects the ride.

*Pre/post tilt.* The pre/post tilt check repeats the rest-window trick after the ride and compares the two gravity vectors. Beyond 25° the user reoriented the phone mid-ride — pulled it out of a pocket, handed it off, flipped it on a table — so

| **Check** | **Condition** | **Threshold** |
|---|---|---|
| *Hard gates* | | |
| Gravity calibration | signed vertical projection available | reject if magnitude-only |
| Phone tilt | $\angle(g_{\text{pre}}, g_{\text{post}}) \leq$ | 25° |
| Ride drift | max gravity drift ≤ | 15° |
| Impact peak | max $\lvert a - \bar{a} \rvert \leq$ | 8 m/s$^2$ |
| Active motion | fraction $\lvert a \rvert > 0.10\,\text{m/s}^2 \geq$ | 0.04 |
| Displacement | $\lvert \widehat{\Delta h} \rvert \in$ | $[0.4, 100]$ m |
| Segment length | active samples $N \geq$ | 30 |
| *Score terms* (penalty sum rejects above 6) | | |
| Pre-window stability | $s_{\text{pre}}$ | graded |
| Pre/post tilt | $\angle(g_{\text{pre}}, g_{\text{post}})$ | graded from 10° |
| Ride drift | max gravity drift | graded |
| Impact peak | max $\lvert a - \bar{a} \rvert$ | graded at 4, 6 m/s$^2$ |
| End-velocity ratio | $\lvert v(T) \rvert / \max \lvert v \rvert$ | graded |

Table 7. ZUPT quality-filter checks and calibrated thresholds.

| **Check** | **Condition** | **Threshold** |
|---|---|---|
| *Hard gates* | | |
| Gravity calibration | signed vertical projection available | reject if magnitude-only |
| Phone tilt | $\angle(g_{\text{pre}}, g_{\text{post}}) \leq$ | 25° |
| Joint $R^2$ floor | $R^2_{\text{joint}} \geq$ (short/mid/long) | 0.35/0.50/0.60 |
| Displacement | $\lvert \widehat{\Delta h} \rvert \in$ | $[0.4, 120]$ m |
| Lobe spacing | $\Delta t_c \geq$ | 0.2 s |
| Active motion | $(2W + \Delta t_c)/T_{\text{ride}} \geq$ | 0.05 |
| Amplitude anchor | $A_{\text{used}}/A_{\text{fit}} \in$ | $[0.5, 2.0]$ |
| Out-of-lobe residual | residual density, outside/inside ≤ | 4 |
| Cruise steadiness | $\text{CV}(v_{\text{cruise}}) \leq$ | 0.6 |
| Segment length | samples $N \geq$ | 30 |
| *Score terms* (penalty sum rejects above 6) | | |
| Pre-window stability | $s_{\text{pre}}$ | graded |
| Pre/post tilt | $\angle(g_{\text{pre}}, g_{\text{post}})$ | graded from 10° |
| Ride drift | max gravity drift | graded |
| Joint $R^2$ | $R^2_{\text{joint}}$ | graded below 0.8 |
| Residual structure | lag-1 residual ACF | graded at 0.25, 0.4, 0.6 |

Table 8. Trapezoid quality-filter checks and calibrated thresholds. The filter accepts a ride only when every hard gate passes and the penalty score stays below its cap.

no single projection serves both ends. Below 10° the check is silent; between 10° and the hard 25° gate it adds a graded penalty.

*Ride drift.* Pre and post agreeing leaves a loophole: the phone can swing in the middle and swing back. Ride drift slices the ride into one-second chunks and records the largest gravity angle relative to the pre-window vector. A held phone stays within a few degrees; an actively handled one climbs past the 15° gate, and the vertical projection becomes ambiguous.

*Motion-content checks (both filters).* These checks reject segments where the algorithms have nothing useful to fit.

*Displacement bounds.* Displacement bounds reject estimates outside the physically plausible range. Below 0.4 m is shorter than one stair step — no real elevator ride registers that small. Above 100 m for ZUPT or 120 m for the trapezoid fit is past the tallest building in the campaign, so the estimate has run away. Either extreme rejects outright.

*Active motion.* The active-motion check rejects segments that carry almost no acceleration signal. ZUPT counts the fraction of samples with $|a| > 0.10\,\mathrm{m/s^2}$ inside the segment and requires at least 4 %. The trapezoid filter asks the equivalent question on the fitted geometry: the two lobes plus the cruise gap together must cover at least 5 % of the ride window. Both hard gates fire only on near-stationary phones, so in practice the graded score terms carry the work.

*Segment length.* Segment length is a structural floor on the noise model. Both error budgets scale with the number of active samples $N$, and below $N = 30$ sample-count variance dominates the per-ride $\sigma$, so the reported interval no longer reflects the noise model. This floor is geometric, not tuned.

*ZUPT-only checks.*

*Impact peak.* The impact-peak gate rejects rides where a non-elevator spike dominates the trace. ZUPT integrates acceleration twice, so any spike feeds straight into $\Delta h$. A pocket drop, a tap, or a hand-off easily clears $8\,\mathrm{m/s^2}$ — ten times the elevator's own lobes. The gate rejects past $8\,\mathrm{m/s^2}$, and the score grades the approach with knees at 4 and $6\,\mathrm{m/s^2}$.

*End-velocity ratio.* The end-velocity ratio scores how cleanly the ZUPT integrator returned to rest. A genuine ride starts and ends at zero velocity, so $|v(T)|/\max|v|$ stays near zero on a healthy fit, and a large ratio signals integrator drift the cruise plateau cannot absorb. The magnitude-detrended trace pins the ratio near zero by construction, so this check stays a graded score term, never a hard gate.

*Trapezoid fit-quality checks.*

*Joint $R^2$.* The joint $R^2$ floor measures how well the trapezoid template explains the trace. Short rides carry less acceleration energy and have intrinsically worse signal-to-noise, so the floor scales with predicted displacement: 0.35 for short, 0.50 for mid, 0.60 for long rides. A fit below the matching floor never modelled the ride.

*Residual autocorrelation.* Residual autocorrelation catches what $R^2$ misses: a template that captures the lobe energy yet leaves a slow, structured bump in the residual — an unmodelled disturbance the squared-error sum cannot see. The lag-1 autocorrelation of the residual flags that structure. Smoothing precedes the fit, so adjacent residuals are correlated by construction even on a perfect ride; the score knees therefore sit high, at 0.25, 0.40, and 0.60.

*Trapezoid fit-geometry checks.*

*Amplitude anchor ratio.* The amplitude anchor ratio compares the two amplitudes the pipeline produces for the same trapezoid: $A_{\mathrm{fit}}$ from the matched filter and $A_{\mathrm{used}}$ from the integrated cruise velocity. A trustworthy fit needs little rescaling between them, so a ratio outside $[0.5,\ 2.0]$ flags a template that locked onto the wrong scale — typically a matched-filter false positive with the wrong $(W, f)$.

*Out-of-lobe residual.* Out-of-lobe residual density flags mid-ride disturbances the trapezoid template missed. A clean fit concentrates structure inside the two lobe windows, so the residual energy density outside should be no higher than inside. A density ratio past fourfold means something leaked energy into the cruise gap that the template never modelled — a passenger moving, a phone re-gripped, a multi-stop trip miscounted as one ride.

*Cruise steadiness.* Cruise steadiness verifies the cabin reached constant velocity between the two lobes. The ZUPT-integrated velocity in that window holds a flat plateau on a healthy ride. A coefficient of variation past 0.6 means the plateau wobbled, so the cabin never settled into steady cruise — a short hop, a stop-go trip, or a misaligned lobe pair.

*Lobe spacing ($\Delta t_c$).* Lobe spacing rejects pairs too close to be a take-off and a landing. A centroid spacing below 0.2 s means the matched filter paired two halves of the same lobe rather than two real lobes, so the closed-form $\Delta h$ in (27) collapses to a single-lobe area the model does not interpret.

## E Evaluation: metrics and supporting figures

This appendix supplies the metric definitions and the diagnostic figures behind the headline tables of §5.

### E.1 Metric definitions

Every column of Tables 1, 2, and 3 comes from one of the metrics in Table 9. Each row pairs the metric with its physical meaning and with why the headline tables include it.

| Metric | Definition | Why we report it |
|---|---|---|
| $F1^*$ | Composite F1 over the four failure modes (`clean`, `missed`, `merged`, `split`, FP); clean/(clean + $\frac{1}{2}$ $\text{bad}_{\text{GT}}$ + $\frac{1}{2}$ $\text{bad}_{\text{pred}}$). | Distinguishes a miss-heavy configuration from a merge-heavy one; a pooled IoU-F1 averages the two. |
| IoU-F1@0.5 | PASCAL-style F1 at intersection-over-union ≥ 0.5. | Comparable to any temporal-detection baseline; depressed by the GT pad of Figure 10. |
| Median (m) | Median of $\lvert\widehat{\Delta h} - \Delta h\rvert$ over the ride set. | Robust scale of the bulk; the right read on "typical" per-ride accuracy. |
| MAE (m) | Mean of $\lvert\widehat{\Delta h} - \Delta h\rvert$. | Outlier-sensitive; the gap between median and MAE is the long-tail diagnostic. |
| ≤ 1.5 m | Fraction of rides with absolute error ≤ 1.5 m. | One-floor hit rate at the dataset's ∼ 3 m inter-floor pitch (§B.3) — the right floor on either side. |
| ≤ 3 m | Fraction of rides with absolute error ≤ 3 m. | Two-floor hit rate; bounds the worst miss a downstream floor-detector still recovers. |
| Coverage | Fraction of rides with $\lvert\widehat{\Delta h} - \Delta h\rvert \le k\sigma$; nominal 90 %. | Empirical check of the conformal guarantee (32). |
| Median CI (m) | Median of $k\sigma$ across the ride set. | Typical interval width; the user's read on how tight a "90 % CI" actually is. |
| Accept rate | Fraction of predictions the quality filter passes. | Recall trade-off of the filter; complements the per-ride scores. |

Table 9. Every metric used in the headline evaluation tables, with the definition and the reason the contract needs it. Coverage and Median CI together carry the calibrated-interval promise of §1; the ≤ 1.5 m and ≤ 3 m columns translate that promise into the floor-detection unit the downstream consumer uses.

### E.2 Gyroscope reconstruction ablation

Table 10 isolates the gain from the earth-frame reconstruction of §2.6: it scores both predictors on the same pooled clean rides with the gyroscope off (`none`, the fixed rest-frame gravity projection) and on (`Mahony`). Reconstruction cuts the mean $\Delta h$ error 1.8–2.3× and the outlier-driven RMSE 4–6×; the RMSE gain is larger because the worst damage without it falls on the few heavily rotated rides, whose fixed-gravity tilt bias runs to tens of metres. The one-floor rate rises on both predictors, most for ZUPT, whose double integration compounds any leaked tilt.

| Algorithm | Reconstruction | MAE (m) | RMSE (m) | $\leq 1.5$ m | $\leq 3$ m |
|---|---|---|---|---|---|
| ZUPT | none (baseline) | 2.38 | 19.14 | 77.6% | 88.9% |
| ZUPT | **Mahony** | **1.05** | **3.28** | **87.9%** | **92.5%** |
| Trapezoid | none (baseline) | 1.61 | 11.49 | 87.5% | 91.3% |
| Trapezoid | **Mahony** | **0.91** | **2.76** | **89.4%** | **93.9%** |

Table 10. Prediction accuracy without (none) and with (Mahony) the accel+gyro earth-frame reconstruction, pooled across train and test (586 clean rides). Reconstruction cuts the mean error 1.8–2.3× and the RMSE tail 4–6× (ZUPT 2.38 ⟶ 1.05 m MAE, 19.1 ⟶ 3.3 m RMSE; trapezoid 1.61 ⟶ 0.91 m, 11.5 ⟶ 2.8 m); the Mahony CIs stay conformally calibrated to 90 % (Table 2).

## E.3 Segmentation supporting figures

Figure 21 disaggregates Table 1 into its five failure categories. The `clean` bar dominates both splits; the `missed` and `FP` tails are shorter on the held-out test than on the tuned train pool.

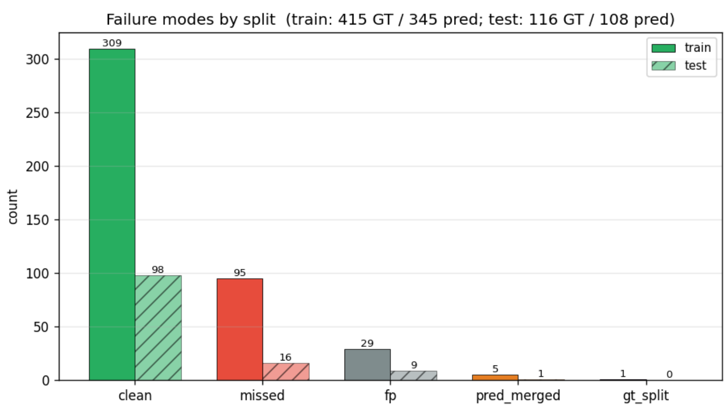



Fig. 21. Segmentation failure-mode counts on the train pool (solid) and the held-out test building (hatched), normalised by GT ride count. The bulk of the train pool's 95 missed rides comes from three heavily damped Pixel 10 sessions where the matched-filter amplitude floor swallows the lobes (§6); the held-out building carries no such session, driving the test-set lift on $F1^*$.

## E.4 Per-ride prediction supporting figures

Two diagnostics anchor Table 2: the empirical CDF that produces the $\leq 1.5$ m and $\leq 3$ m columns, and the signed-error PDF that checks for bias.

*Error CDF.*. Figure 22 plots the cumulative distribution of $|\widehat{\Delta h} - \Delta h|$ for the trapezoid fit across the pooled clean rides. The $\leq 1.5$ m and $\leq 3$ m fractions of Table 2 are the $y$-values where the curve crosses the two vertical lines.

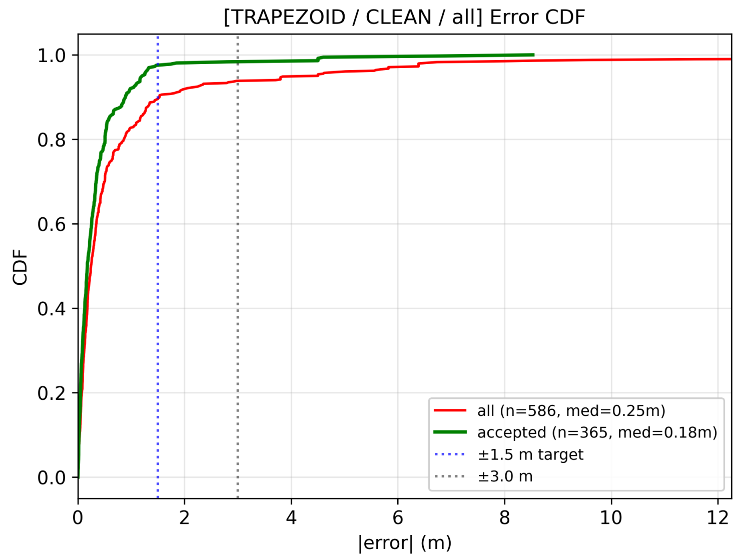



Fig. 22. Empirical CDF of absolute $\Delta h$ error for the trapezoid fit on clean rides (all rides, red; quality-accepted, green). Vertical dashed lines mark the ±1.5 m and ±3 m floor targets; the all-ride curve crosses ±1.5 m by roughly the 89 % percentile and ±3 m by the 94 % percentile, while the accepted curve clears ±1.5 m by the 97 % percentile.

*Signed-error distribution.* The signed-error PDF (Figure 23) checks that the predictor is not systematically biased upward or downward. A symmetric, zero-centred PDF means the fit's bias is small compared to its variance; any drift in the mean would otherwise corrupt the calibrated interval of §4.2.4.


Fig. 23. Density of signed error $\widehat{\Delta h} - \Delta h$ on pooled clean trapezoid rides. The PDF is centred near zero; the tails are heavier than Gaussian, which is why the distribution-free conformal interval of (32) is the right calibration tool rather than a $1.645\,\sigma$ Gaussian quantile.

*Per-view scatter.* Figure 24 stitches the trapezoid fit, the matched-segmenter view, and the all-segments-versus-barometer view onto one panel each. The off-diagonal tail in the right panel is the false-positive cost the quality filter of §5.3 removes.

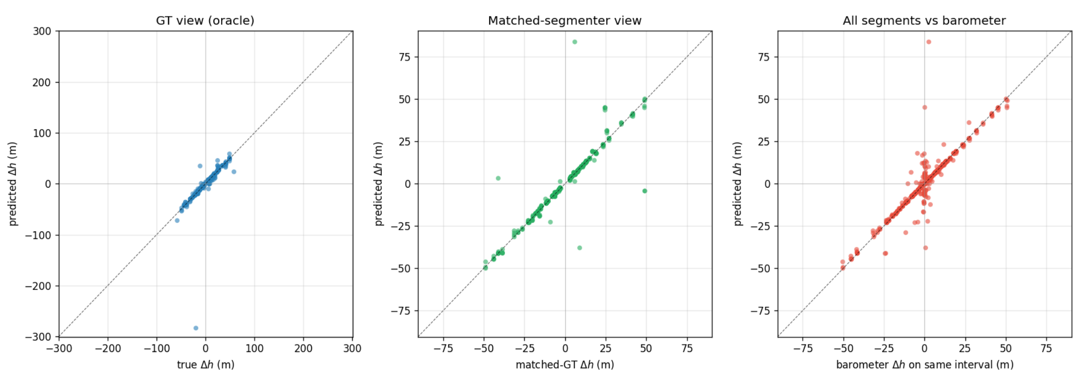



Fig. 24. Predicted versus ground-truth $\Delta h$ on every clean ride, one panel per evaluation view: GT-segments (predictor on every GT interval, left), matched-segmenter (predictor on segmenter intervals that cleanly match a GT, middle), and all-segments-versus-barometer (predictor on every segmenter interval scored against the barometer, right). Mean error in Table 2 is driven by the far-from-diagonal points, where double integration of a body-frame trace accumulates drift on the damped recordings.

## E.5 Pipeline supporting figures

Figure 25 shows the deployed pipeline's accepted-only CDF: the same axis as Figure 22 after the quality filter is in the loop. Every view crosses the $\pm 1.5$ m floor target before the 90th percentile.

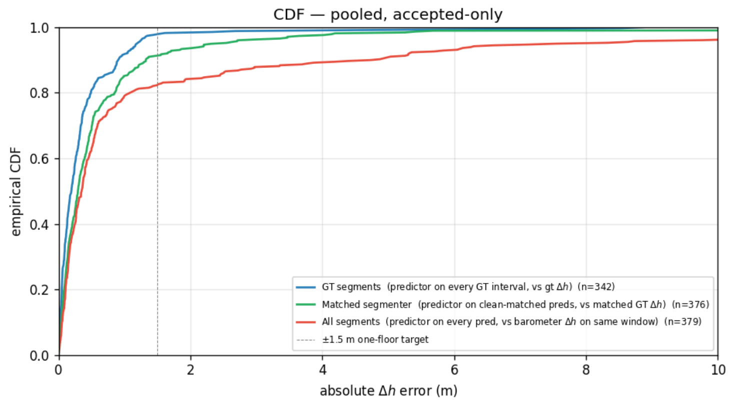



Fig. 25. Empirical CDF of absolute $\Delta h$ error on the deployed pipeline (accepted-only), pooled across both splits and broken into the three evaluation views of Figure 24. The trade-off the filter introduces is recall: 18 % of clean trapezoid rides and 7 % of clean ZUPT rides are rejected, so a downstream consumer sees about four out of every five real rides — each flagged with a coverage-guaranteed interval.